

twoday

Introduction to AI-native development

Agentic engineering – agents that answers, participates, and acts in software development



Assistant and interactive AI

An assistant can summarize a customer case.

Autonomous and process AI

An agent can inspect the case, retrieve relevant information, evaluate it against policy, prepare the next action and perhaps execute part of the process.

Agentic engineering – agents that answers, participates, and acts in software development

AI that **answers** and **participate**

Assistant and interactive AI

An assistant can summarize a customer case.

AI that **participates** and **acts**

Autonomous and process AI

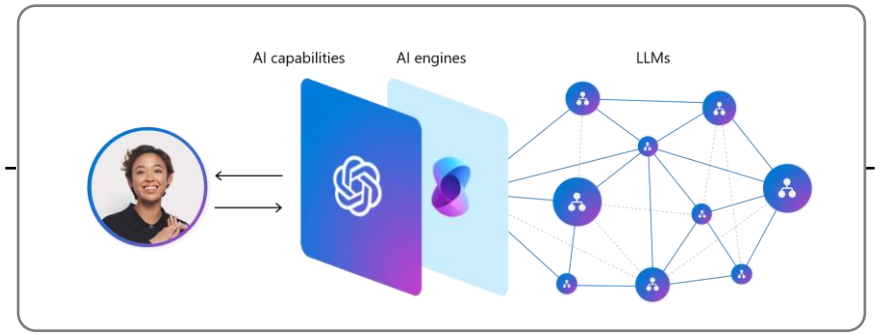
An agent can inspect the case, retrieve relevant information, evaluate it against policy, prepare the next action and perhaps execute part of the process.

Agentic **Engineering**

These agents are specialized for **software development** tasks. They range from assisted coding to **automated development** workflows, integrating AI directly into how software is built and maintained.

Agentic capabilities versus agentic engineering

Agentic **capabilities**

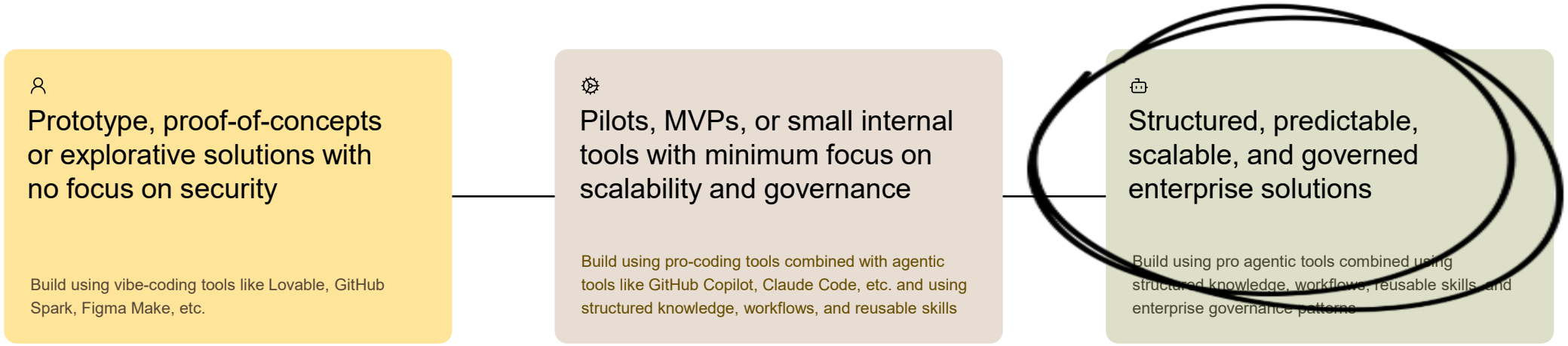


Agentic **Engineering**

Enabling AI and agentic features and capabilities

Applying Agentic Development Life Cycle

Different levels of agentic engineering



Not recommended for production but great for testing ideas, working on different workflow, A/B testing.

Can be done by most people in the organization without any technical knowledge including knowledge about enterprise security, governance, etc.

Ready for production but with limited focus on security, governance, and scalability.

Build using structured knowledge, reusable components and skills.

Development is handled by agents but validated by developers.

Optimal for small to medium greenfield and brownfield projects.

Ready for production with extensive focus on security, governance, and scalability.

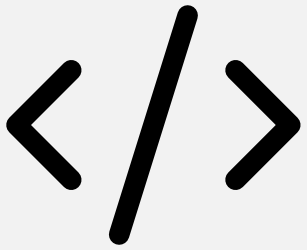
Build using structured knowledge, workflows, reusable components, and skills.

Development is handled by agents but validated by developers.

Optimal for medium to large greenfield and brownfield projects with mixed tech stack including modernization.

How development changes

MANUAL CODING



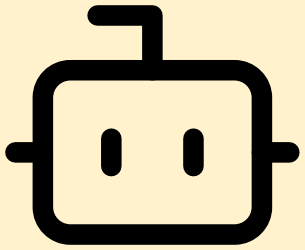
Developer writes everything

Manual scaffolding

Hand-written tests

Large code surface

AI-ASSISTED DEVELOPMENT



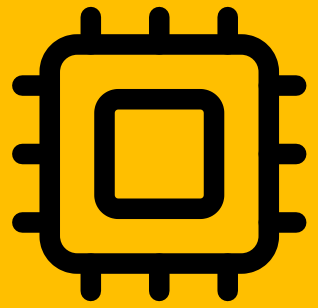
Developer writes the code

AI code suggestions

Prompt-driven scaffolding

AI-assisted refactoring

AI-NATIVE DEVELOPMENT



Developer defines intent

Spec-driven planning

Agent execution

Human validation gates

Before

Manual coding

Producer → Intent designer + reviewer

Now (+)

Agentic Engineering

Evolution in how we engineer AI systems

Prompt
Engineering

Prompt

Designing clear instructions that guide the model toward the intended outcome.

Context
Engineering

Context

Providing the right information, tools, memory, and constraints at the right time.

So much for ‘trust but verify’: Nearly half of software developers don’t check AI-generated code – and 38% say it's because it takes longer than reviewing code produced by colleagues

A concerning number of developers are failing to check AI-generated code, exposing enterprises to huge security threats

The Register

AI coding tools make developers slower but they think they're faster, study finds

Predicted a 24% boost, but clocked a 19% drag

Thomas Claburn

Fri 11 Jul 2025 // 22:41 UTC

Using AI might actually slow down experienced devs

News By Craig Hale published July 11, 2025

Experienced developers aren't getting any benefit from AI, report claims

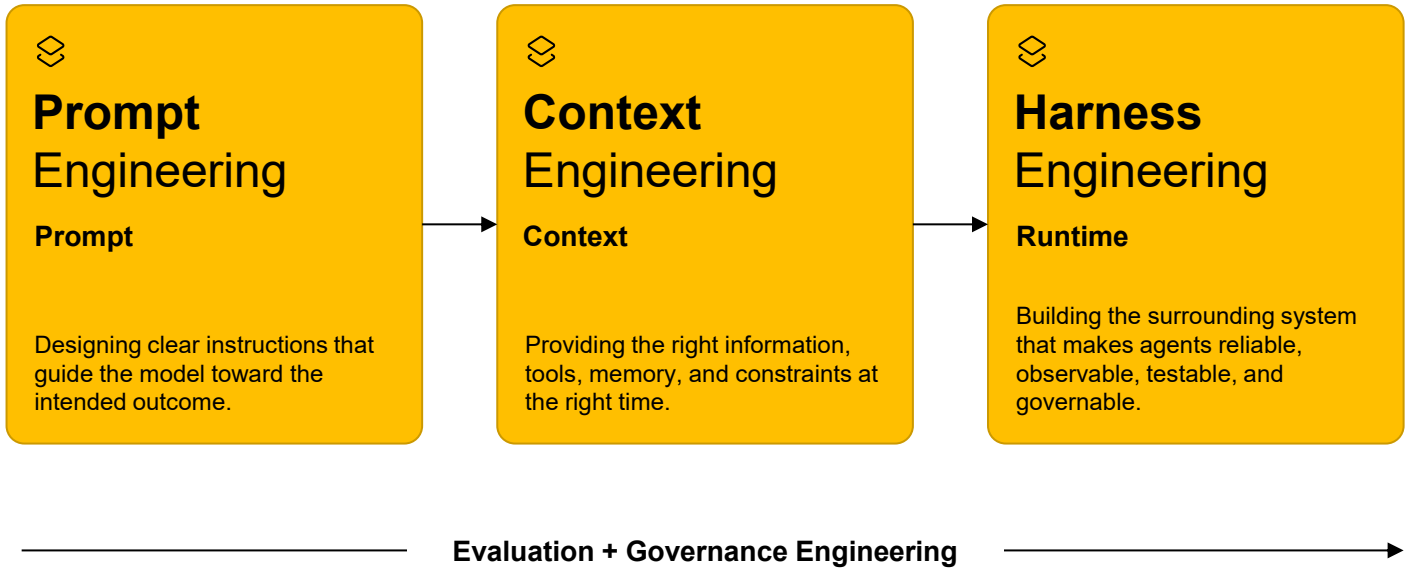
Developers Lean on AI More, But Report Growing Doubts About Accuracy, Stack Overflow Survey Says

The Risks of Using AI in Software Development

Why AI Coding Tools Make Experienced Developers 19% Slower and How to Fix It

Oct 3, 2025 Molisha Shah

Evolution in how we engineer AI systems



Deliver using agent-driven development

Structured, AI-orchestrated processes where the developer defines intent, specifications guide planning, agents execute tasks, and human validation gates ensure control and quality across the end-to-end flow.
Designed to be predictable, reusable, team-scalable, and governed.

DEVELOPMENT FLOW

Getting started

Enable agent-driven development structure and flow to software project.

Structuring knowledge

Describe infrastructure, audience, security model, tech stack, architecture.

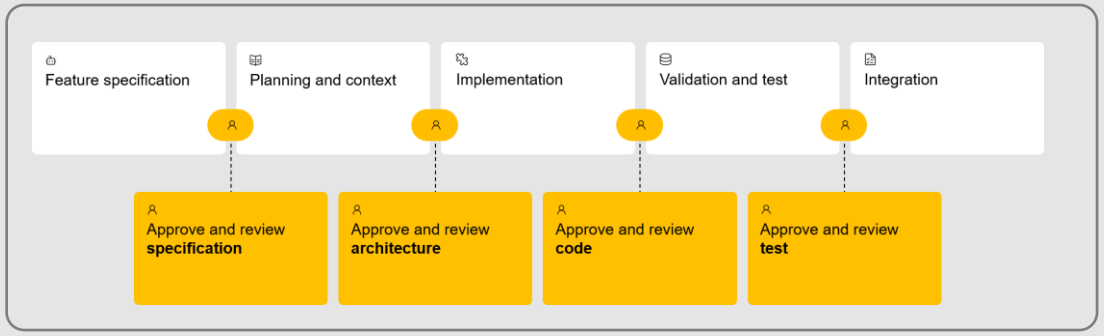
Engineering primitives

Add agents, instructions, skills, and prompts to the solution.

Compile

Spec-driven development

Use spec engine to work on features and tasks and follow spec-driven principals.



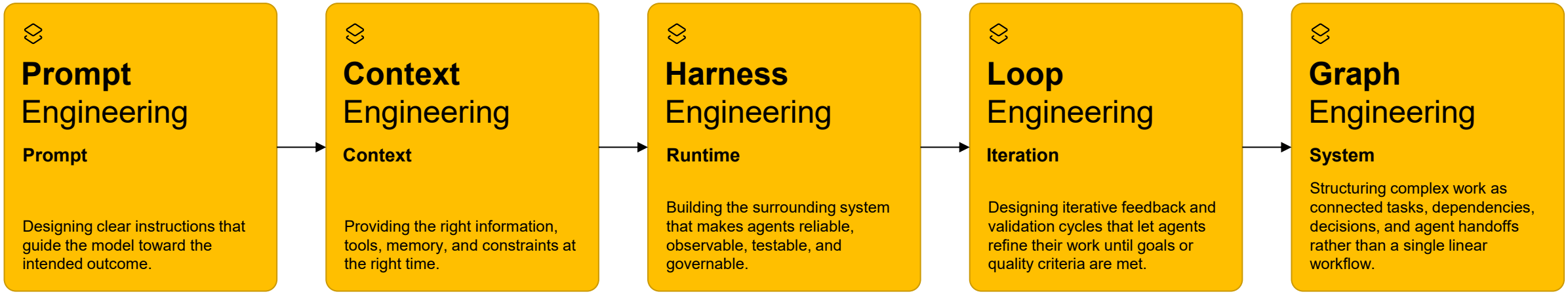
Compile into standard formats

Compile knowledge and primitives into standard formatted artifacts.

Orchestrate and review changes

Use human orchestration and review, and AI test and review

Evolution in how we engineer AI systems



Evolution in how we engineer AI systems

Prompt Engineering

Prompt

Designing clear instructions that guide the model toward the intended outcome.

Context Engineering

Context

Providing the right information, tools, memory, and constraints at the right time.

Harness Engineering

Runtime

Building the surrounding system that makes agents reliable, observable, testable, and governable.

Loop Engineering

Iteration

Designing iterative feedback and validation cycles that let agents refine their work until goals or quality criteria are met.

Graph Engineering

System

Structuring complex work as connected tasks, dependencies, decisions, and agent handoffs rather than a single linear workflow.

Evaluation + Governance Engineering

Welcome to the agenda for **this hackathon**



Predictable, reusable, and scale across teams

Development time decrease with
+50%

Quality and security up with
+20%

Improve standardization
>50%



Happy developers
+20%

Walkthrough of the tools

twoday

GitHub Copilot in
VS Code

demo

twoday

Codex agent in
VS Code

demo

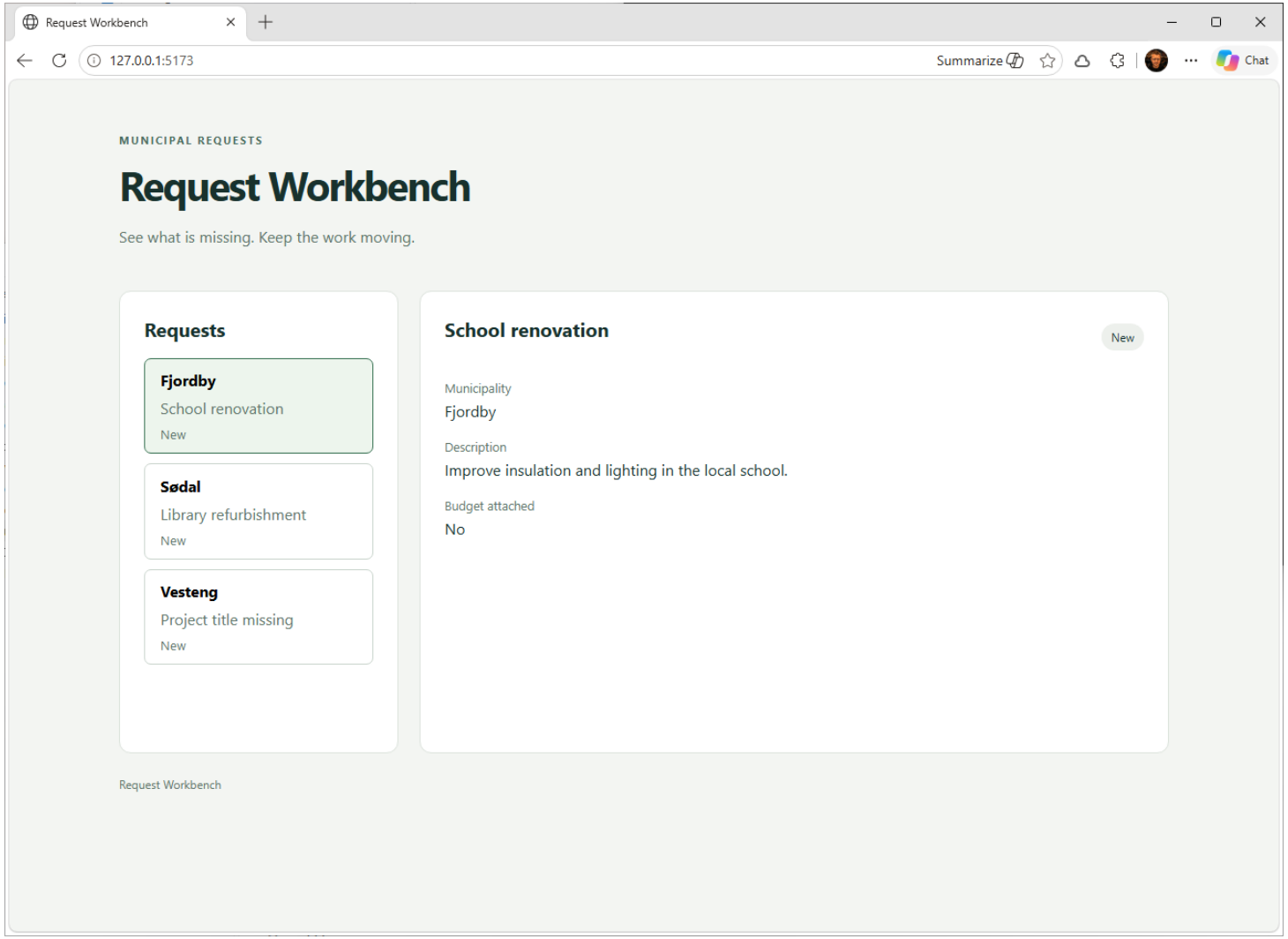
twoday

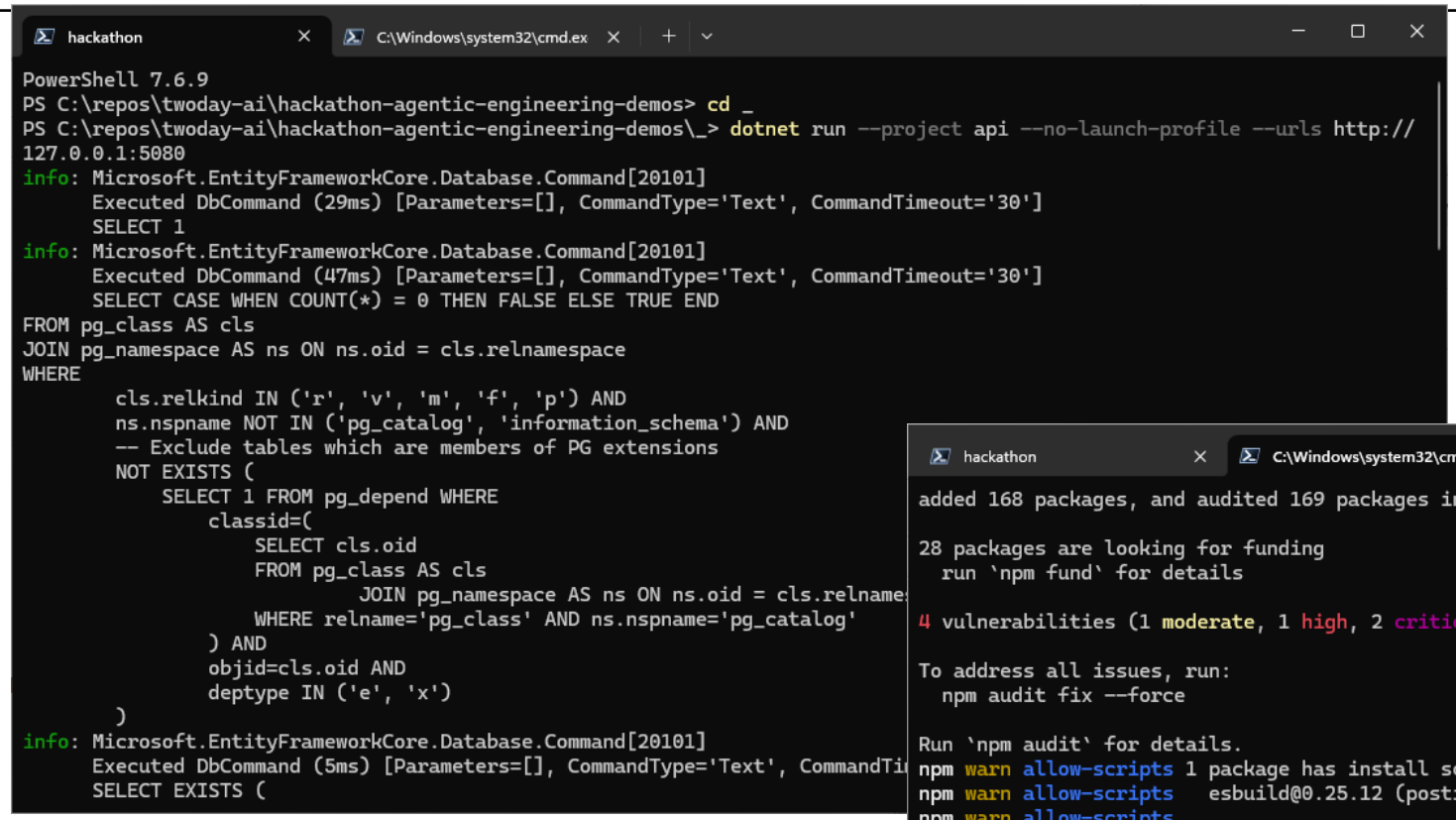
Codex desktop app

demo

AI-native development

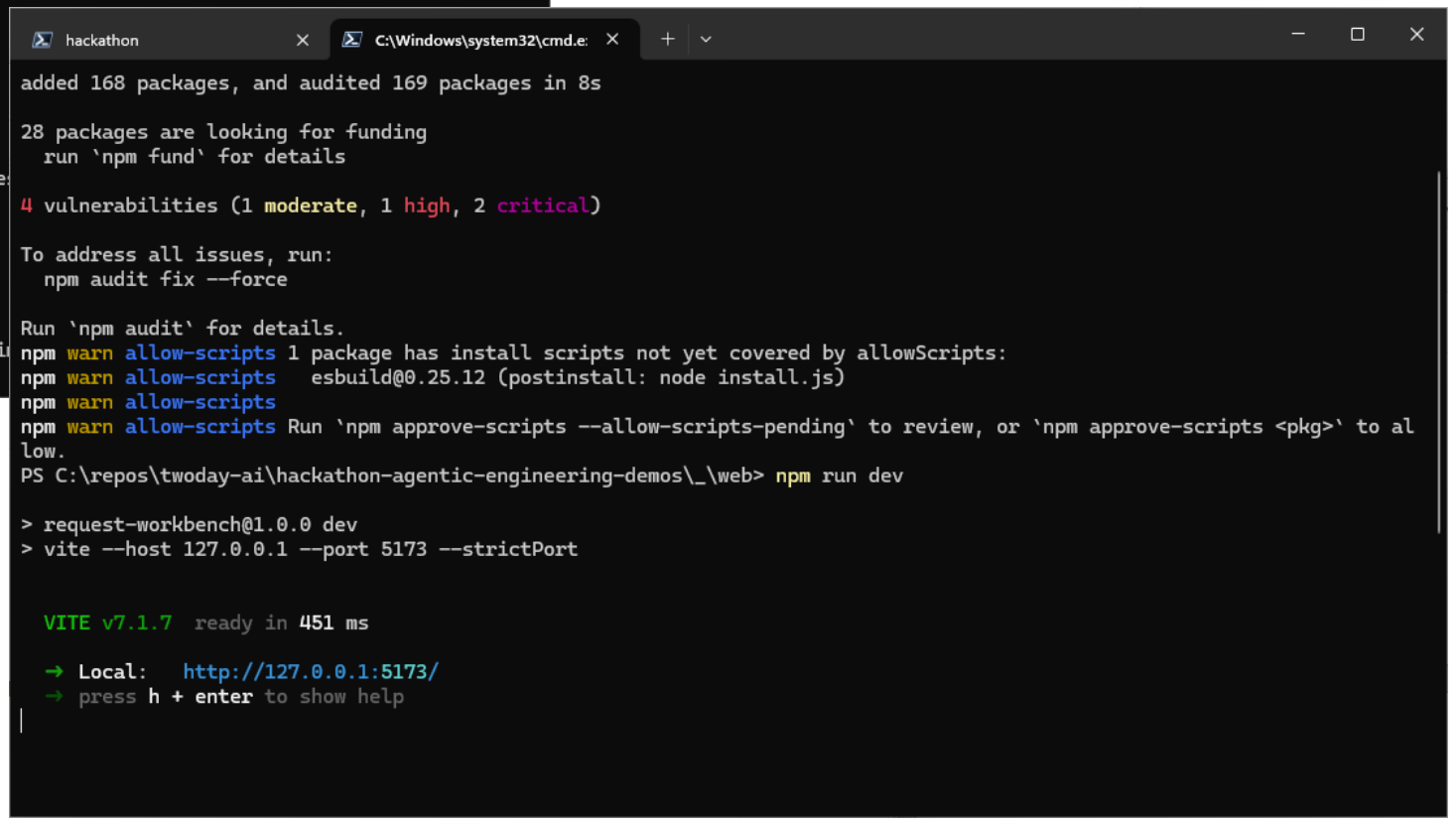
demo


Request Workbench
127.0.0.1:5173
Summarize
Chat
MUNICIPAL REQUESTS
Request Workbench
See what is missing. Keep the work moving.
Requests
Fjordby
School renovation
New
Sødal
Library refurbishment
New
Vesteng
Project title missing
New
School renovation
New
Municipality
Fjordby
Description
Improve insulation and lighting in the local school.
Budget attached
No
Request Workbench


hackathon
C:\Windows\system32\cmd.ex
PowerShell 7.6.9
PS C:\repos\twoday-ai\hackathon-agentic-engineering-demos> cd _
PS C:\repos\twoday-ai\hackathon-agentic-engineering-demos_> dotnet run --project api --no-launch-profile --urls http://
127.0.0.1:5080
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (29ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT 1
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (47ms) [Parameters=[], CommandType='Text', CommandTimeout='30']
SELECT CASE WHEN COUNT(*) = 0 THEN FALSE ELSE TRUE END
FROM pg_class AS cls
JOIN pg_namespace AS ns ON ns.oid = cls.relnamespace
WHERE
cls.relkind IN ('r', 'v', 'm', 'f', 'p') AND
ns.nspname NOT IN ('pg_catalog', 'information_schema') AND
-- Exclude tables which are members of PG extensions
NOT EXISTS (
SELECT 1 FROM pg_depend WHERE
classid=(
SELECT cls.oid
FROM pg_class AS cls
JOIN pg_namespace AS ns ON ns.oid = cls.relname
WHERE relname='pg_class' AND ns.nspname='pg_catalog'
) AND
objid=cls.oid AND
deptype IN ('e', 'x')
)
info: Microsoft.EntityFrameworkCore.Database.Command[20101]
Executed DbCommand (5ms) [Parameters=[], CommandType='Text', CommandTi
SELECT EXISTS (

API

Web

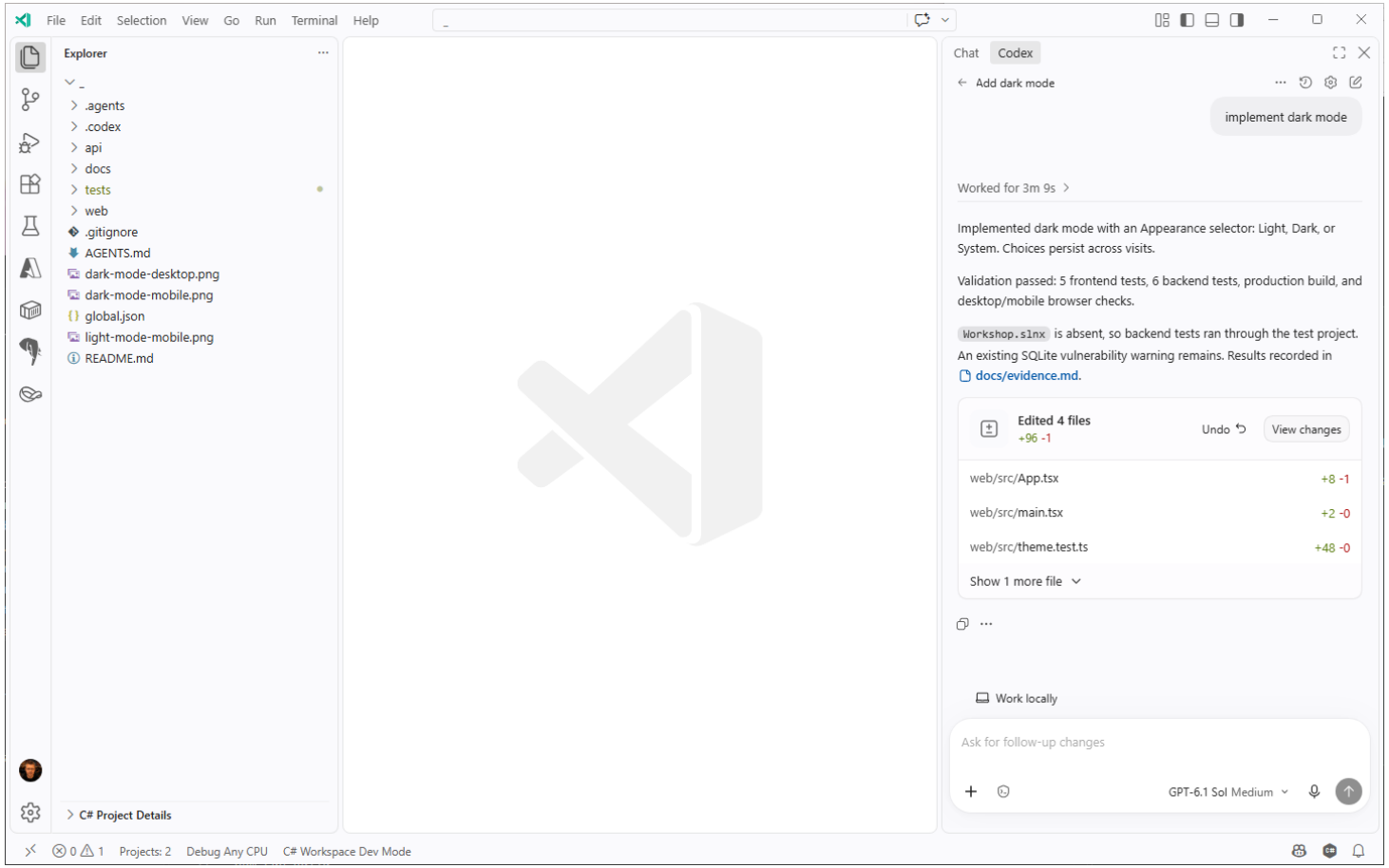

hackathon
C:\Windows\system32\cmd.e:
added 168 packages, and audited 169 packages in 8s
28 packages are looking for funding
run `npm fund` for details
4 vulnerabilities (1 moderate, 1 high, 2 critical)
To address all issues, run:
npm audit fix --force
Run `npm audit` for details.
npm warn allow-scripts 1 package has install scripts not yet covered by allowScripts:
npm warn allow-scripts esbuild@0.25.12 (postinstall: node install.js)
npm warn allow-scripts
npm warn allow-scripts Run `npm approve-scripts --allow-scripts-pending` to review, or `npm approve-scripts <pkg>` to al
low.
PS C:\repos\twoday-ai\hackathon-agentic-engineering-demos_\web> npm run dev
> request-workbench@1.0.0 dev
> vite --host 127.0.0.1 --port 5173 --strictPort
VITE v7.1.7 ready in 451 ms
Local: http://127.0.0.1:5173/
press h + enter to show help

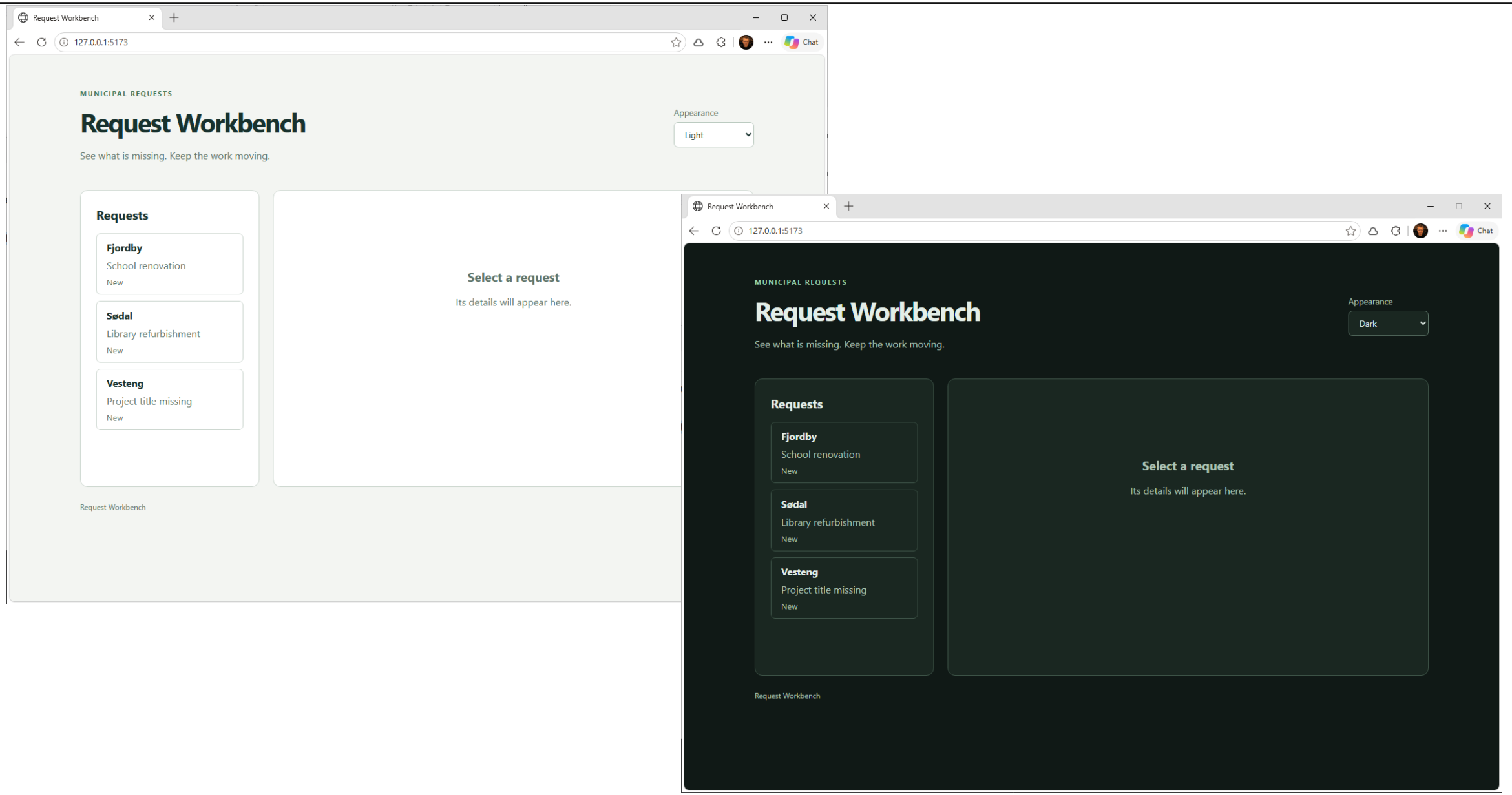
Implement dark mode

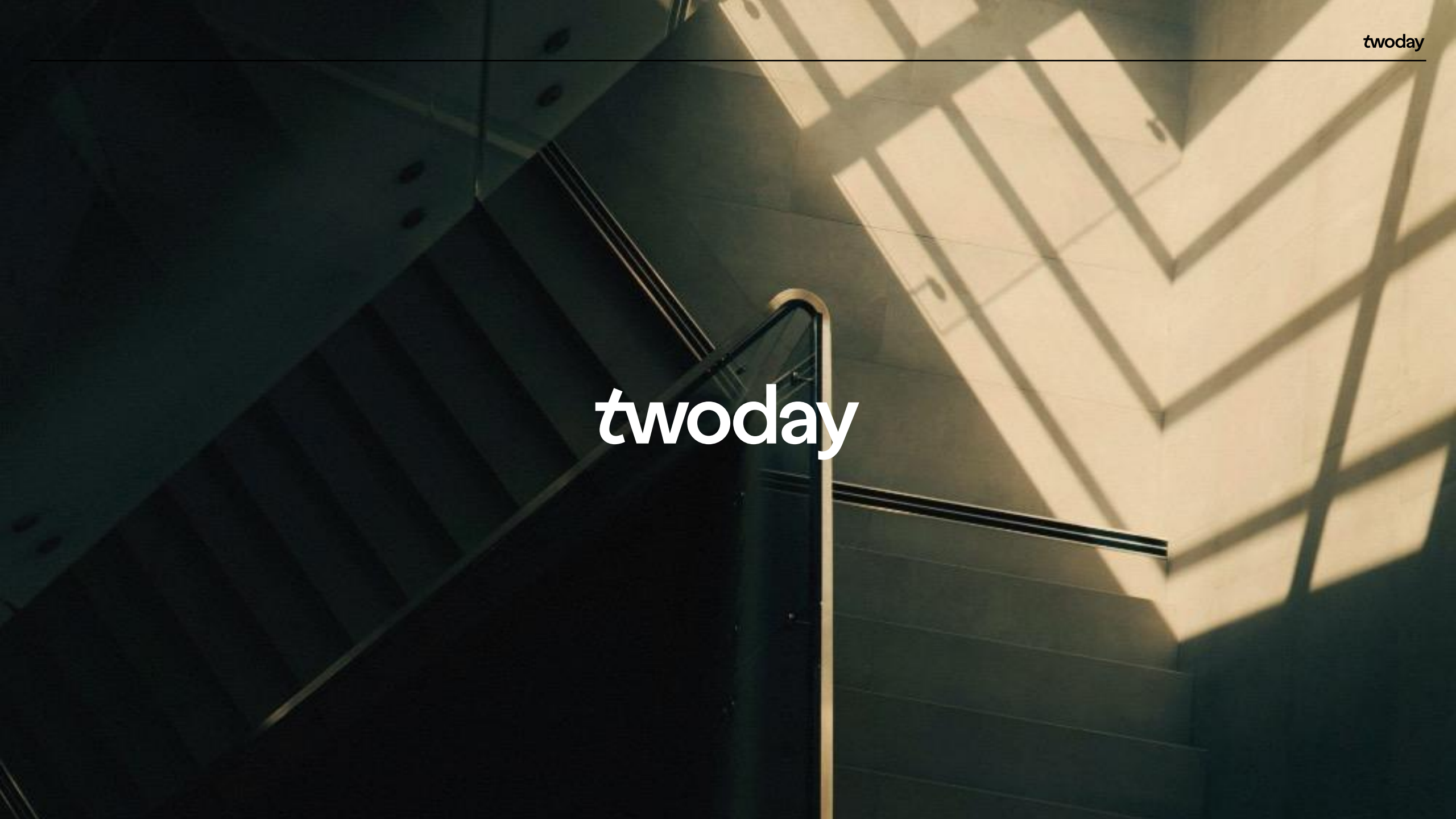
demo

Dark mode

`implement dark mode`

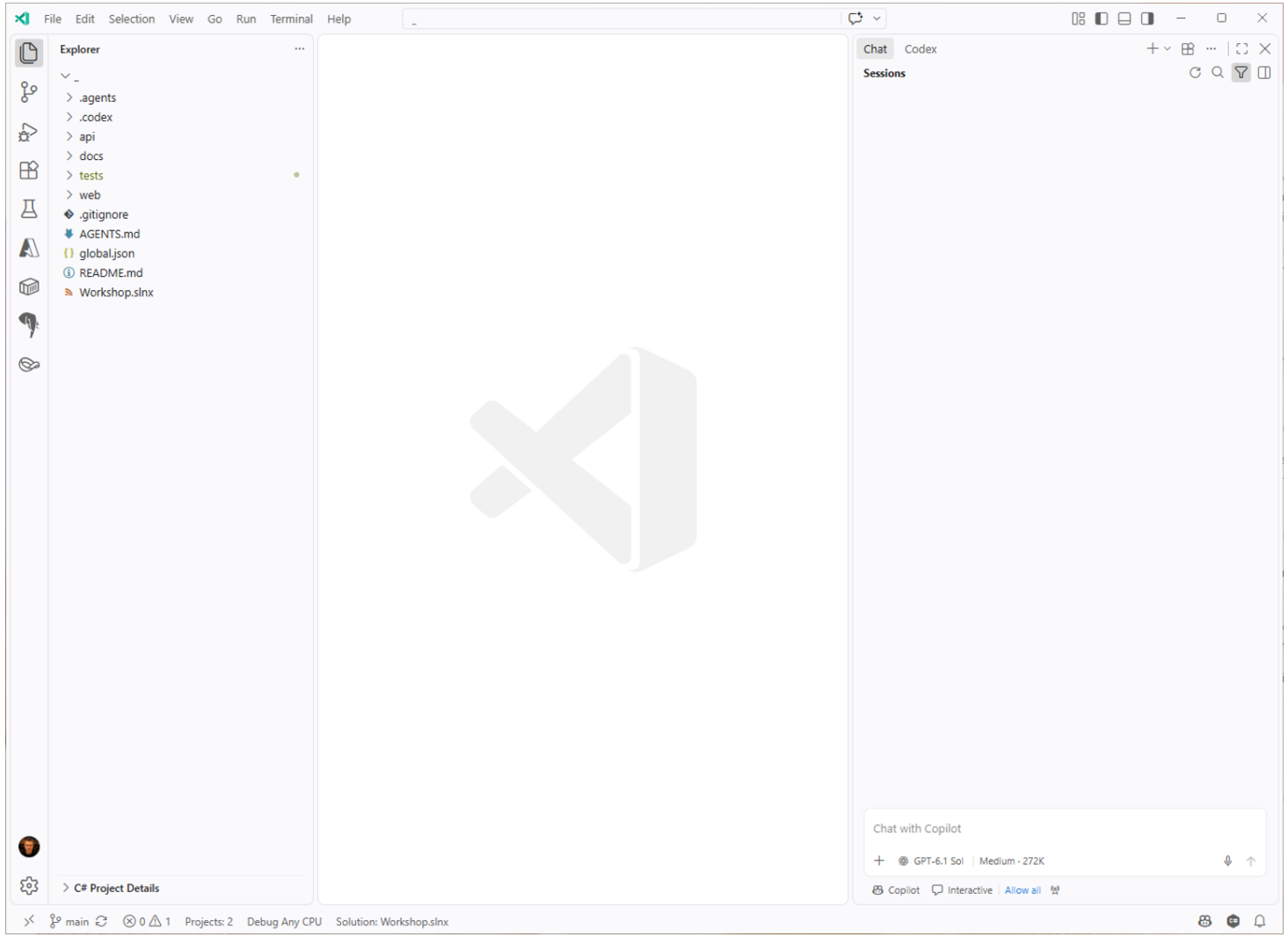

File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
dark-mode-desktop.png
dark-mode-mobile.png
global.json
light-mode-mobile.png
README.md
C# Project Details
Chat
Codex
Add dark mode
implement dark mode
Worked for 3m 9s
Implemented dark mode with an Appearance selector: Light, Dark, or System. Choices persist across visits.
Validation passed: 5 frontend tests, 6 backend tests, production build, and desktop/mobile browser checks.
Workshop.slnx is absent, so backend tests ran through the test project. An existing SQLite vulnerability warning remains. Results recorded in docs/evidence.md.
Edited 4 files
+96 -1
Undo
View changes
web/src/App.tsx +8 -1
web/src/main.tsx +2 -0
web/src/theme.test.ts +48 -0
Show 1 more file
Work locally
Ask for follow-up changes
GPT-6.1 Sol Medium
Projects: 2
Debug Any CPU
C# Workspace Dev Mode

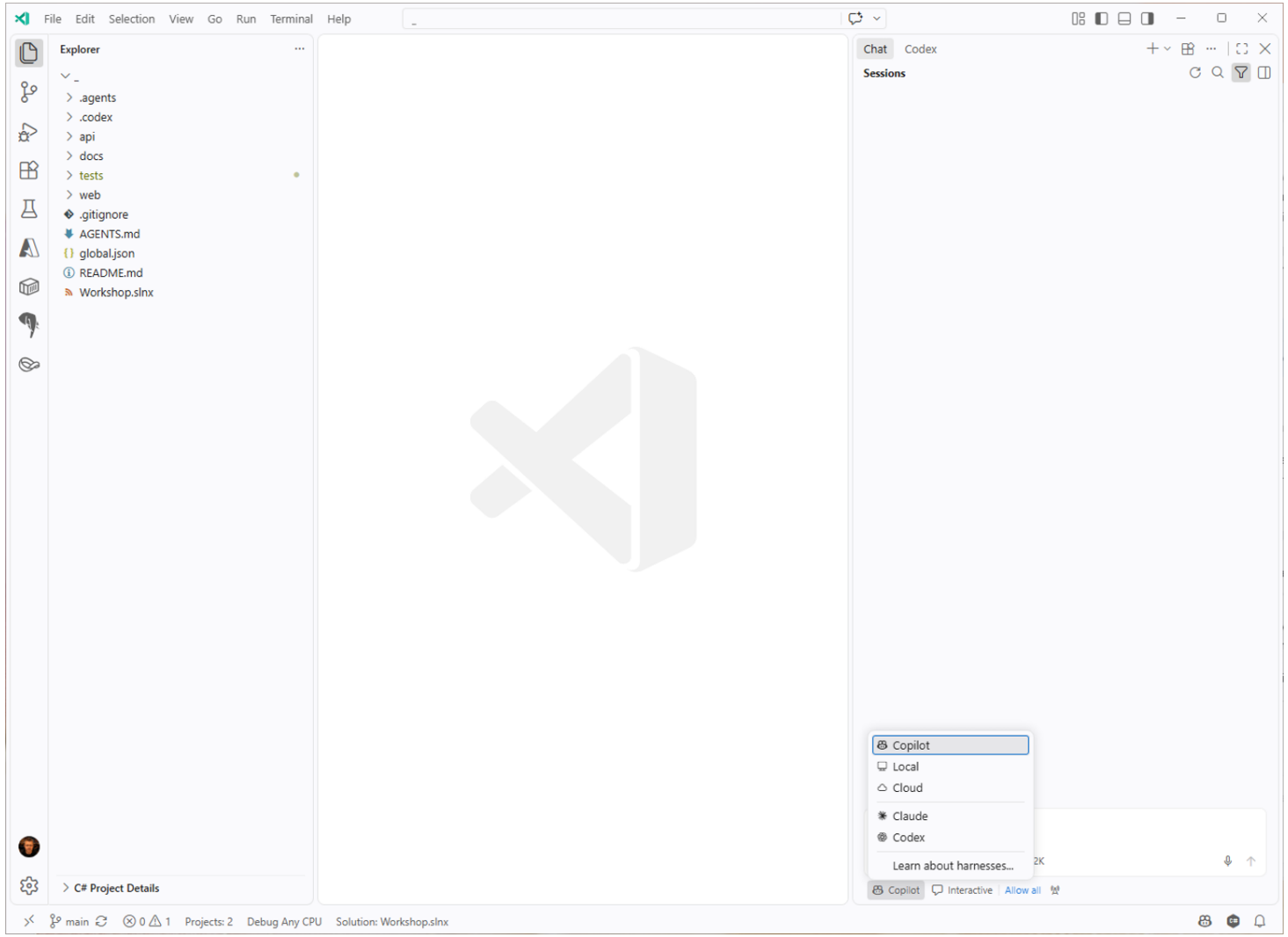

Request Workbench
127.0.0.1:5173
Chat
MUNICIPAL REQUESTS
Request Workbench
See what is missing. Keep the work moving.
Appearance
Light
Requests
Fjordby
School renovation
New
Sødal
Library refurbishment
New
Vesteng
Project title missing
New
Select a request
Its details will appear here.
Request Workbench
Request Workbench
127.0.0.1:5173
Chat
MUNICIPAL REQUESTS
Request Workbench
See what is missing. Keep the work moving.
Appearance
Dark
Requests
Fjordby
School renovation
New
Sødal
Library refurbishment
New
Vesteng
Project title missing
New
Select a request
Its details will appear here.
Request Workbench

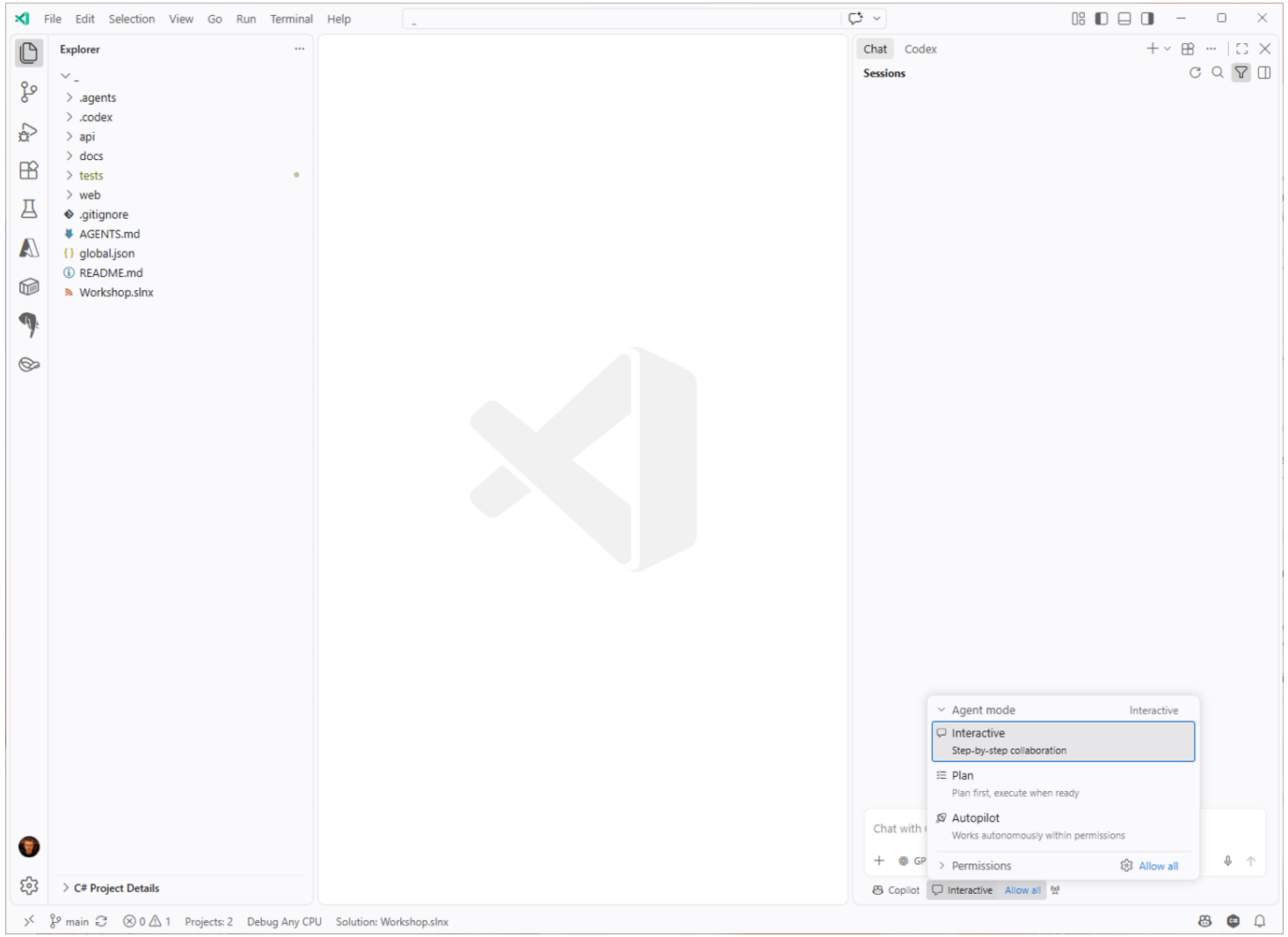

twoday

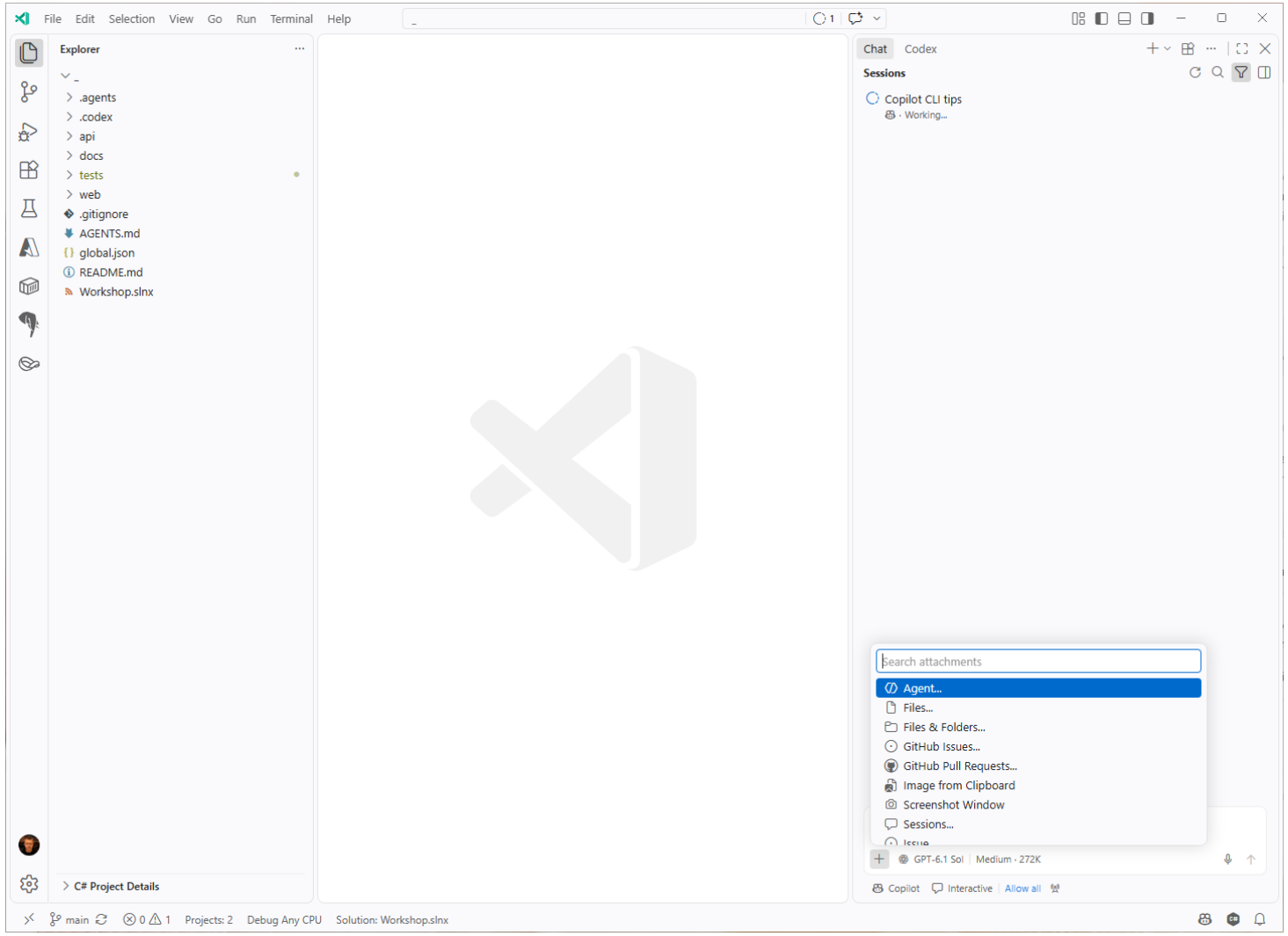
GitHub Copilot in VS Code

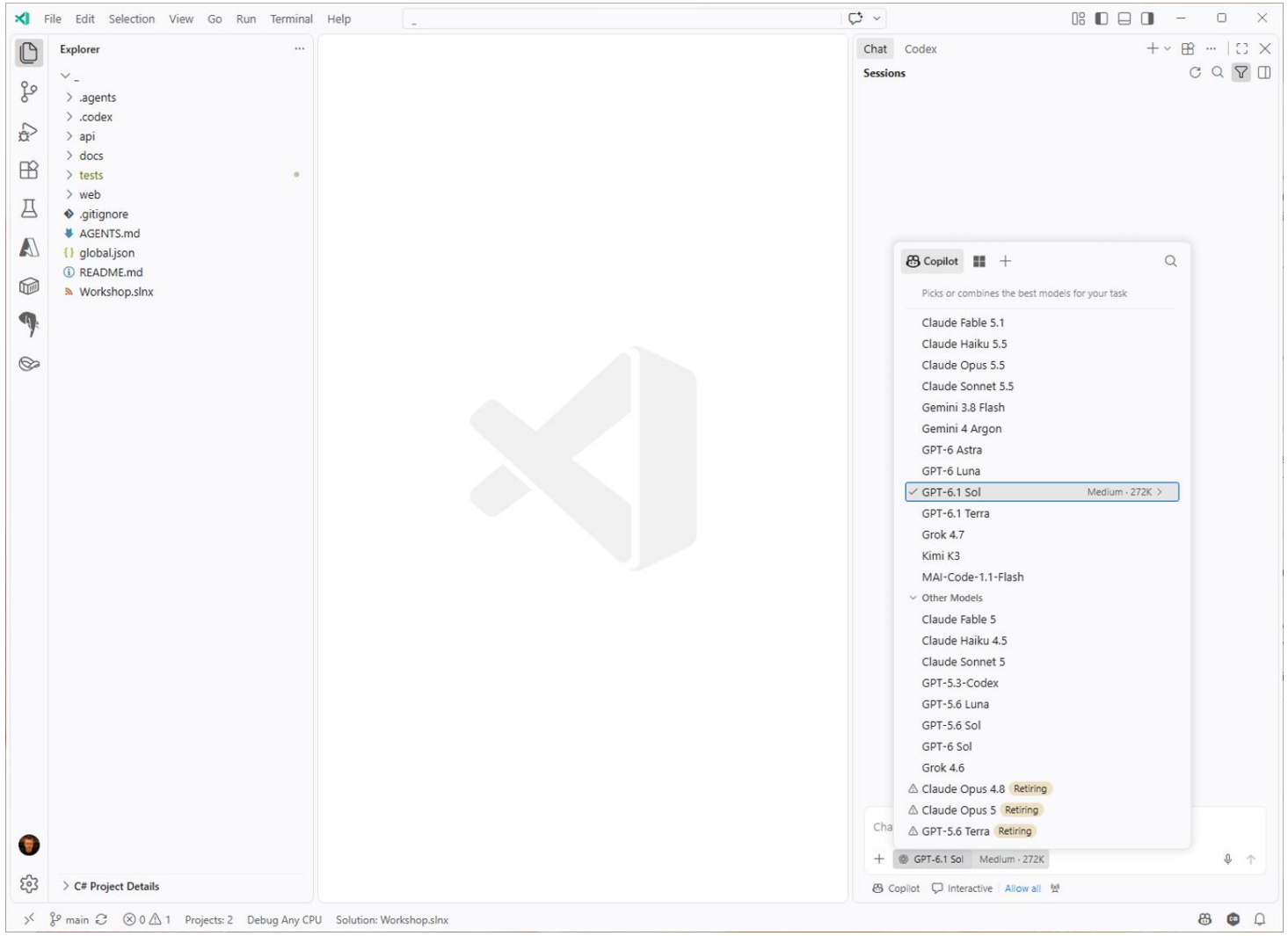
demo

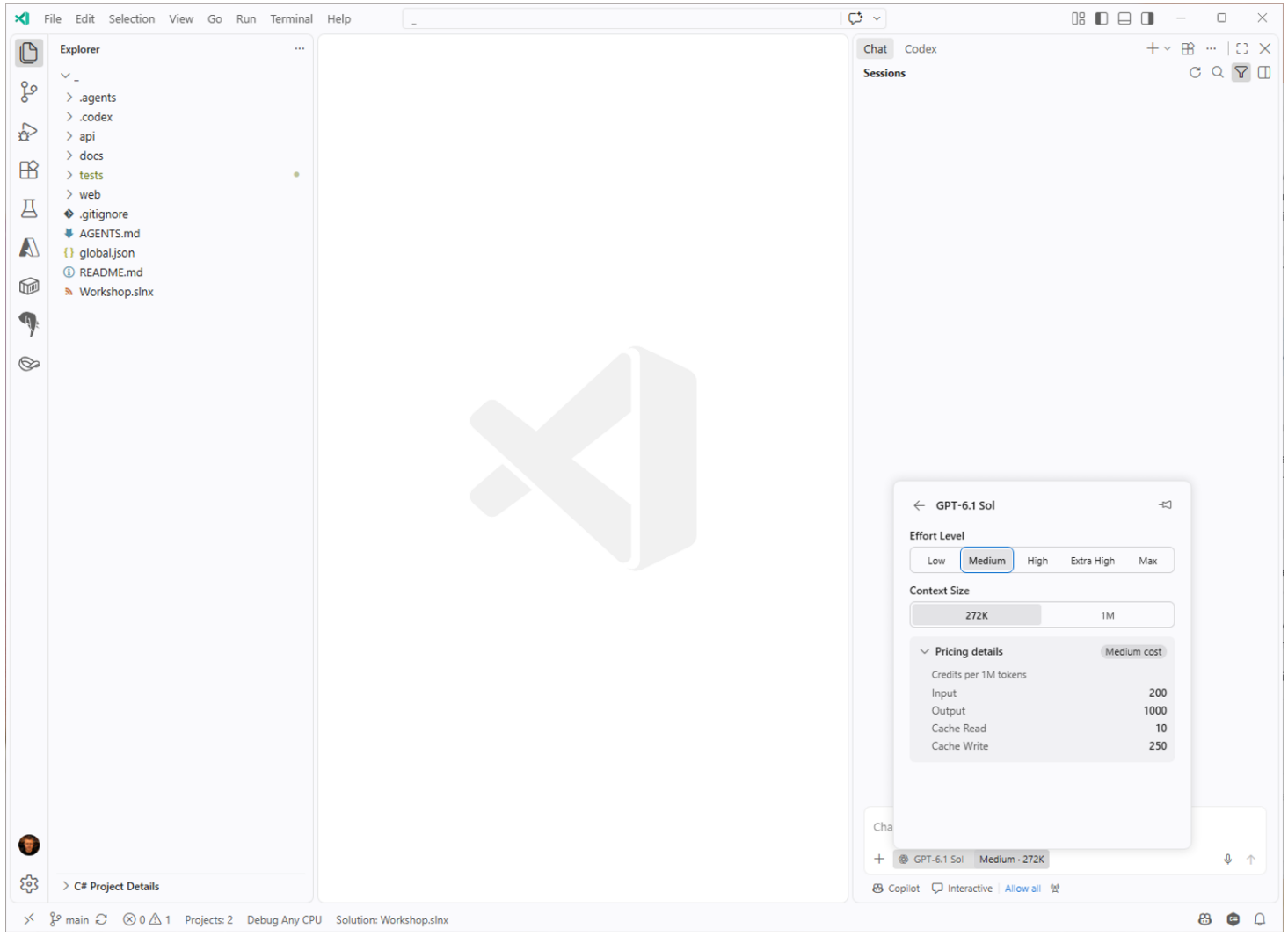

File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Sessions
Chat with Copilot
GPT-6.1 Sol | Medium · 272K
Copilot
Interactive | Allow all
main
0 1
Projects: 2
Debug Any CPU
Solution: Workshop.slnx

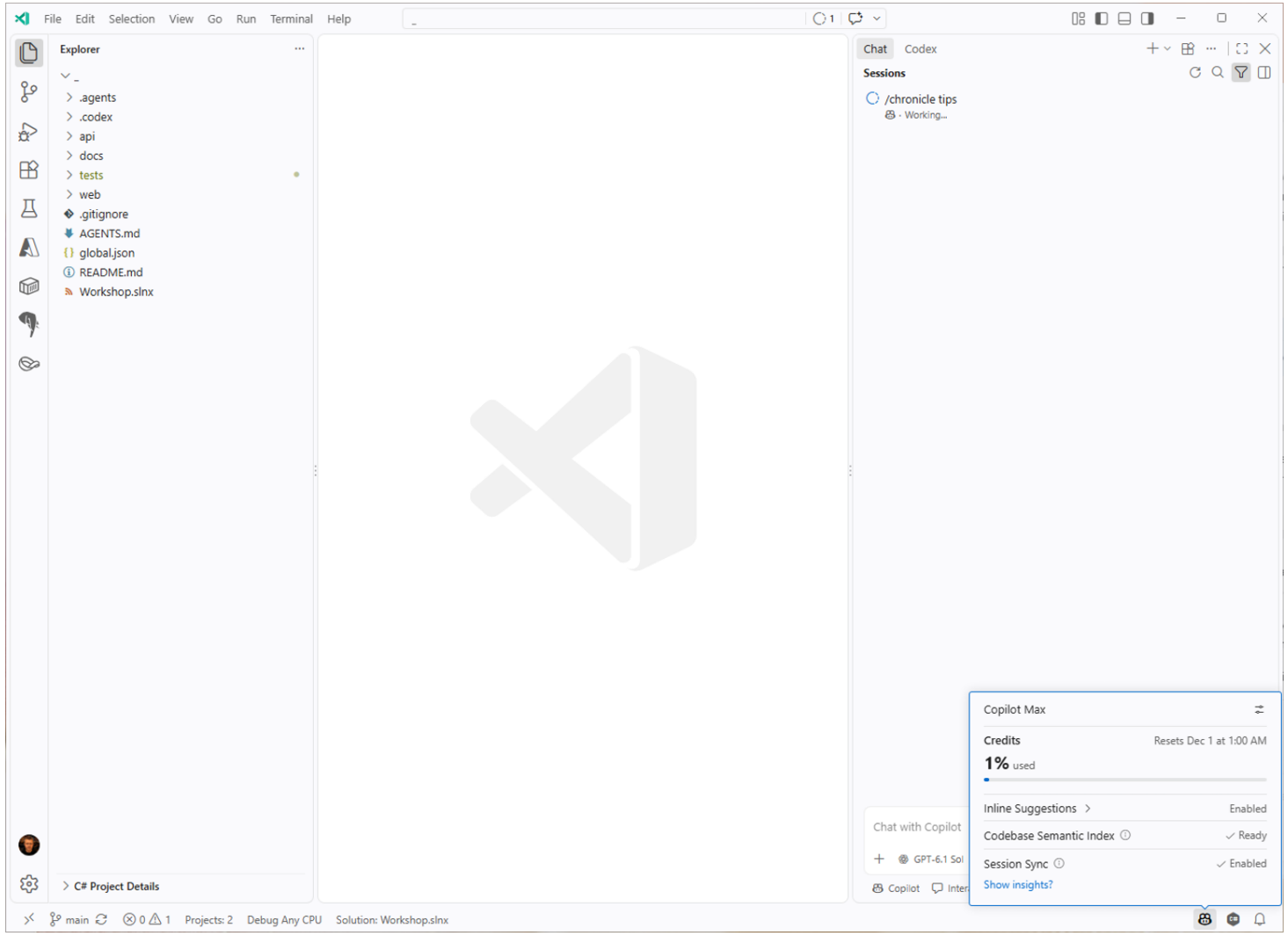

File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Sessions
Copilot
Local
Cloud
Claude
Codex
Learn about harnesses...
Copilot
Interactive
Allow all
main
Projects: 2
Debug Any CPU
Solution: Workshop.slnx


File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Sessions
Agent mode
Interactive
Interactive
Step-by-step collaboration
Plan
Plan first, execute when ready
Autopilot
Works autonomously within permissions
Permissions
Allow all
Chat with
Copilot
Interactive
Allow all
main
0 1
Projects: 2
Debug Any CPU
Solution: Workshop.slnx


File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Sessions
Copilot CLI tips
Working...
Search attachments
Agent...
Files...
Files & Folders...
GitHub Issues...
GitHub Pull Requests...
Image from Clipboard
Screenshot Window
Sessions...
GPT-6.1 Sol | Medium · 272K
Copilot
Interactive
Allow all
main
Projects: 2
Debug Any CPU
Solution: Workshop.slnx

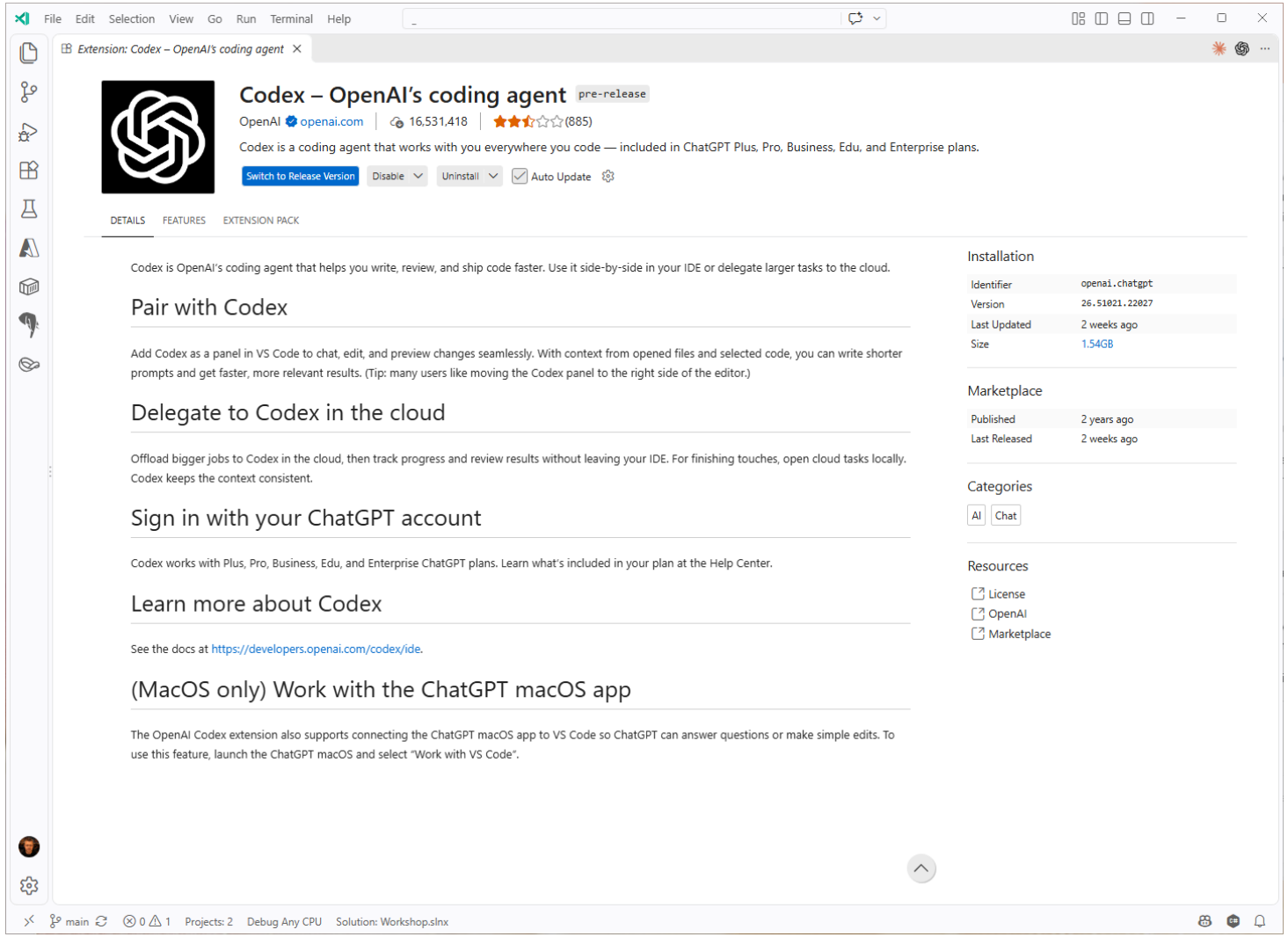

File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Sessions
Copilot
Picks or combines the best models for your task
Claude Fable 5.1
Claude Haiku 5.5
Claude Opus 5.5
Claude Sonnet 5.5
Gemini 3.8 Flash
Gemini 4 Argon
GPT-6 Astra
GPT-6 Luna
GPT-6.1 Sol Medium · 272K
GPT-6.1 Terra
Grok 4.7
Kimi K3
MAI-Code-1.1-Flash
Other Models
Claude Fable 5
Claude Haiku 4.5
Claude Sonnet 5
GPT-5.3-Codex
GPT-5.6 Luna
GPT-5.6 Sol
GPT-6 Sol
Grok 4.6
Claude Opus 4.8 Retiring
Claude Opus 5 Retiring
GPT-5.6 Terra Retiring
GPT-6.1 Sol Medium · 272K
Copilot
Interactive
Allow all
main
Projects: 2
Debug Any CPU
Solution: Workshop.slnx

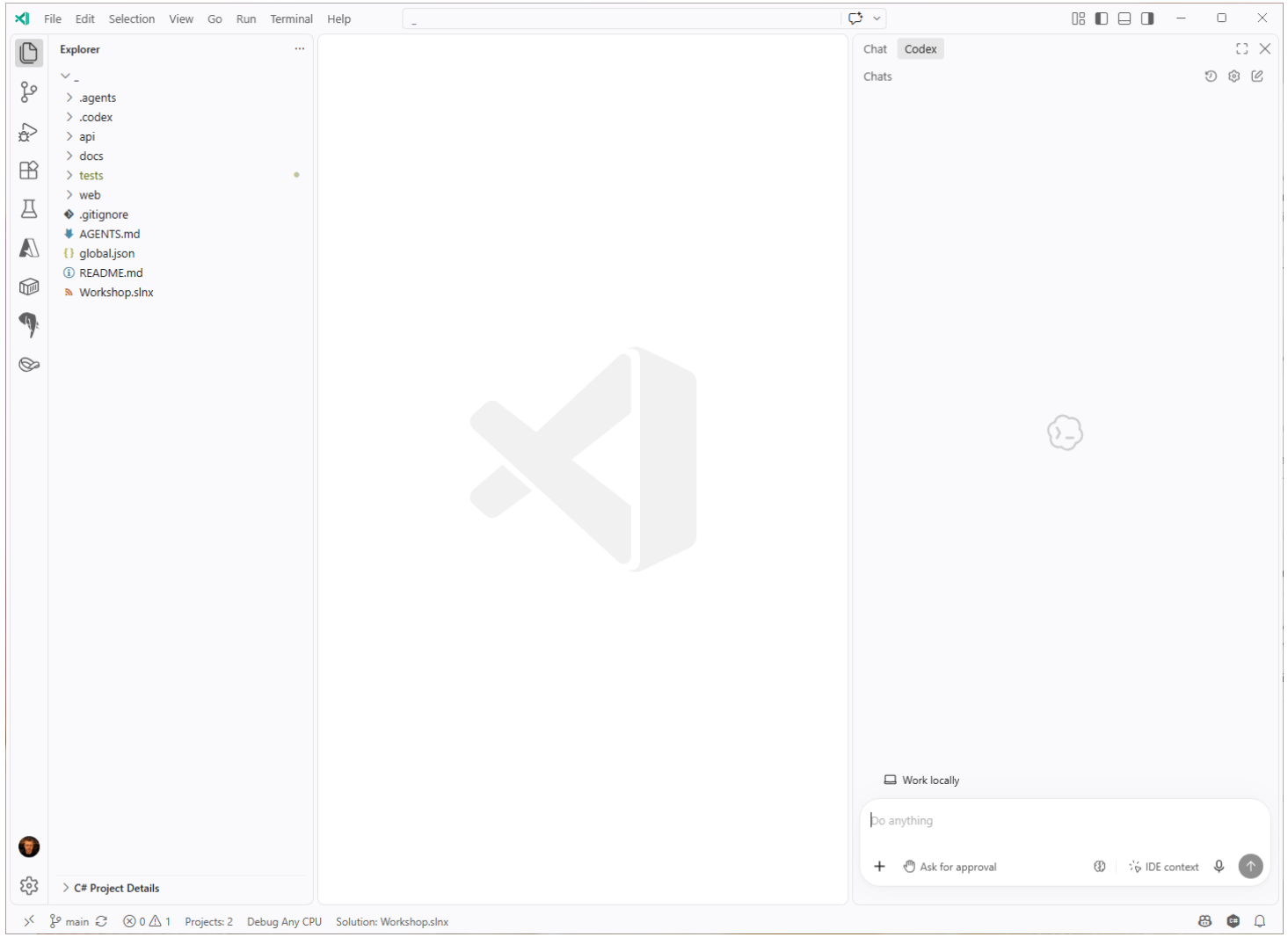

File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Sessions
GPT-6.1 Sol
Effort Level
Low
Medium
High
Extra High
Max
Context Size
272K
1M
Pricing details
Medium cost
Credits per 1M tokens
Input 200
Output 1000
Cache Read 10
Cache Write 250
GPT-6.1 Sol
Medium · 272K
Copilot
Interactive
Allow all
main
Projects: 2
Debug Any CPU
Solution: Workshop.slnx

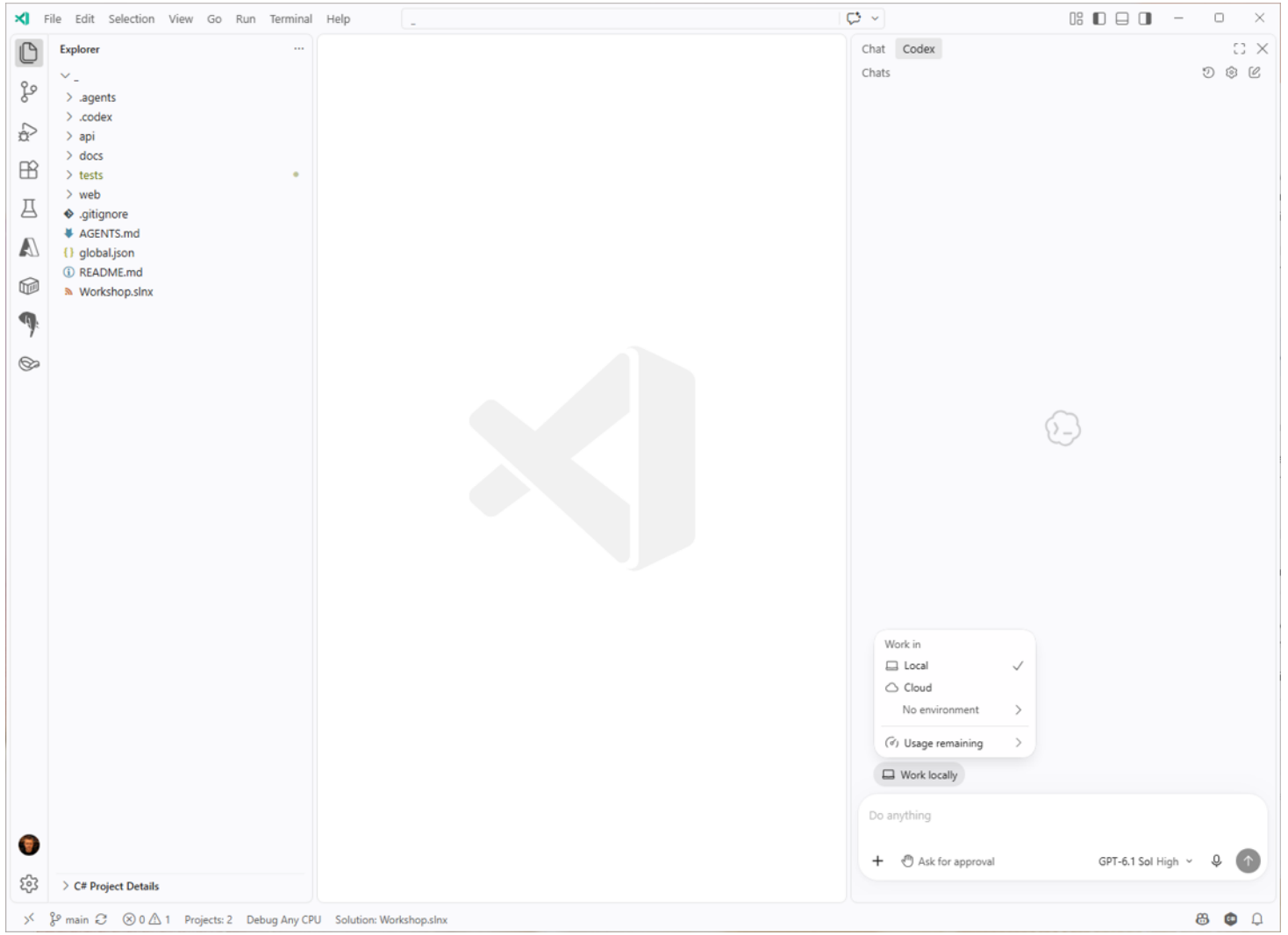

File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Sessions
/chronicle tips
Working...
Chat with Copilot
GPT-6.1 Sol
Copilot
Copilot Max
Credits
Resets Dec 1 at 1:00 AM
1% used
Inline Suggestions
Enabled
Codebase Semantic Index
Ready
Session Sync
Enabled
Show insights?
main
Projects: 2
Debug Any CPU
Solution: Workshop.slnx

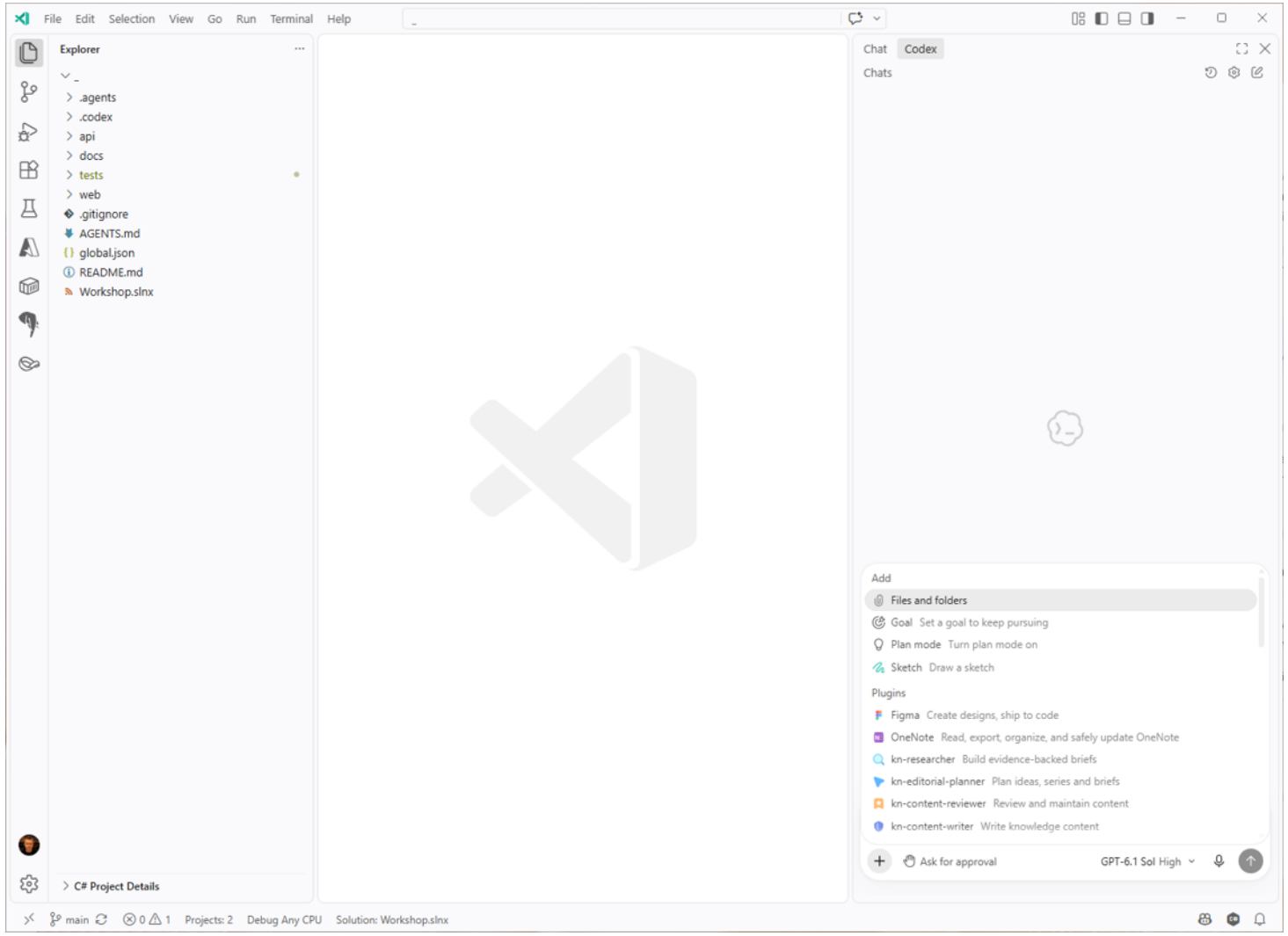
Codex agent in VS Code

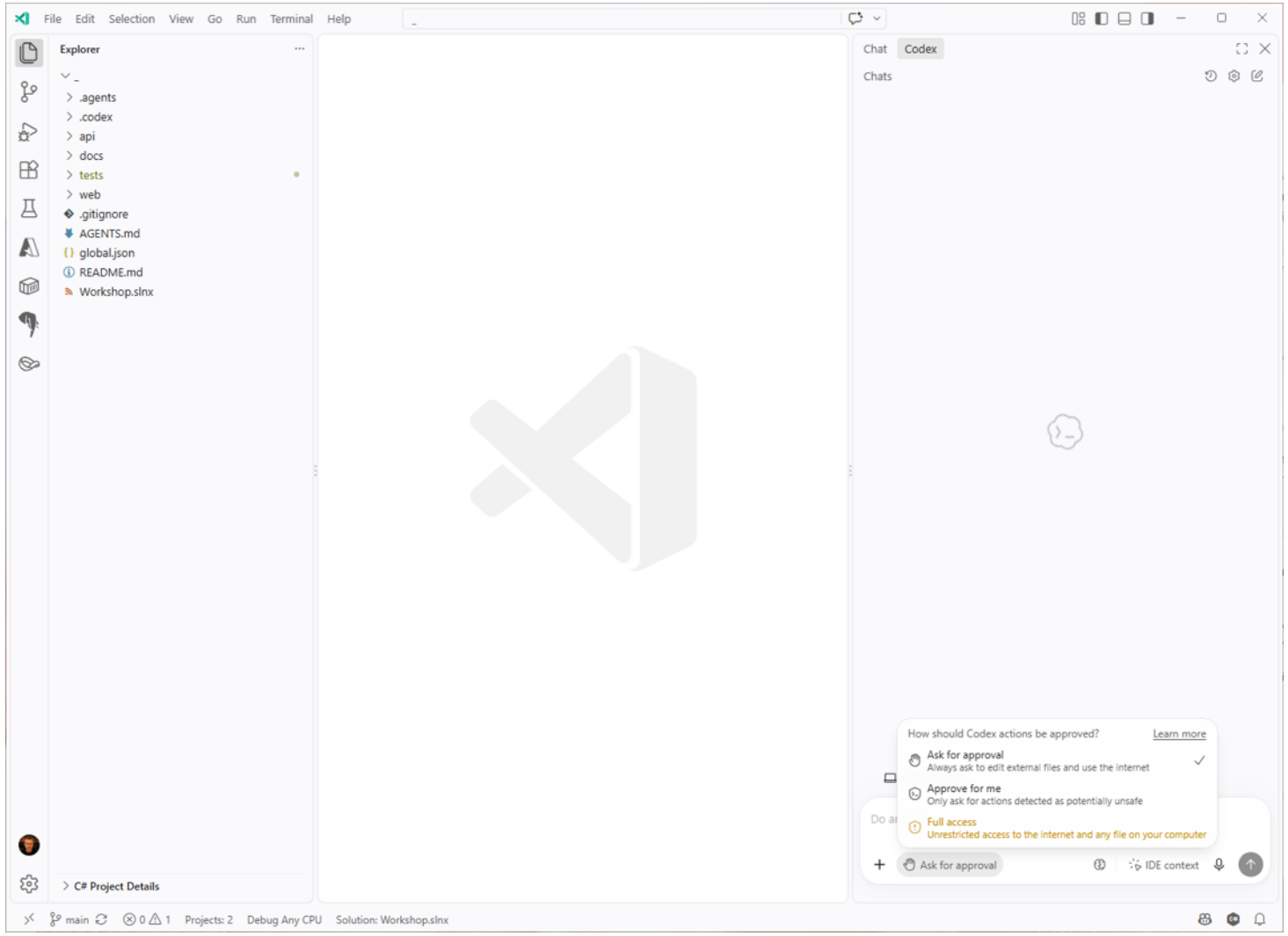
demo

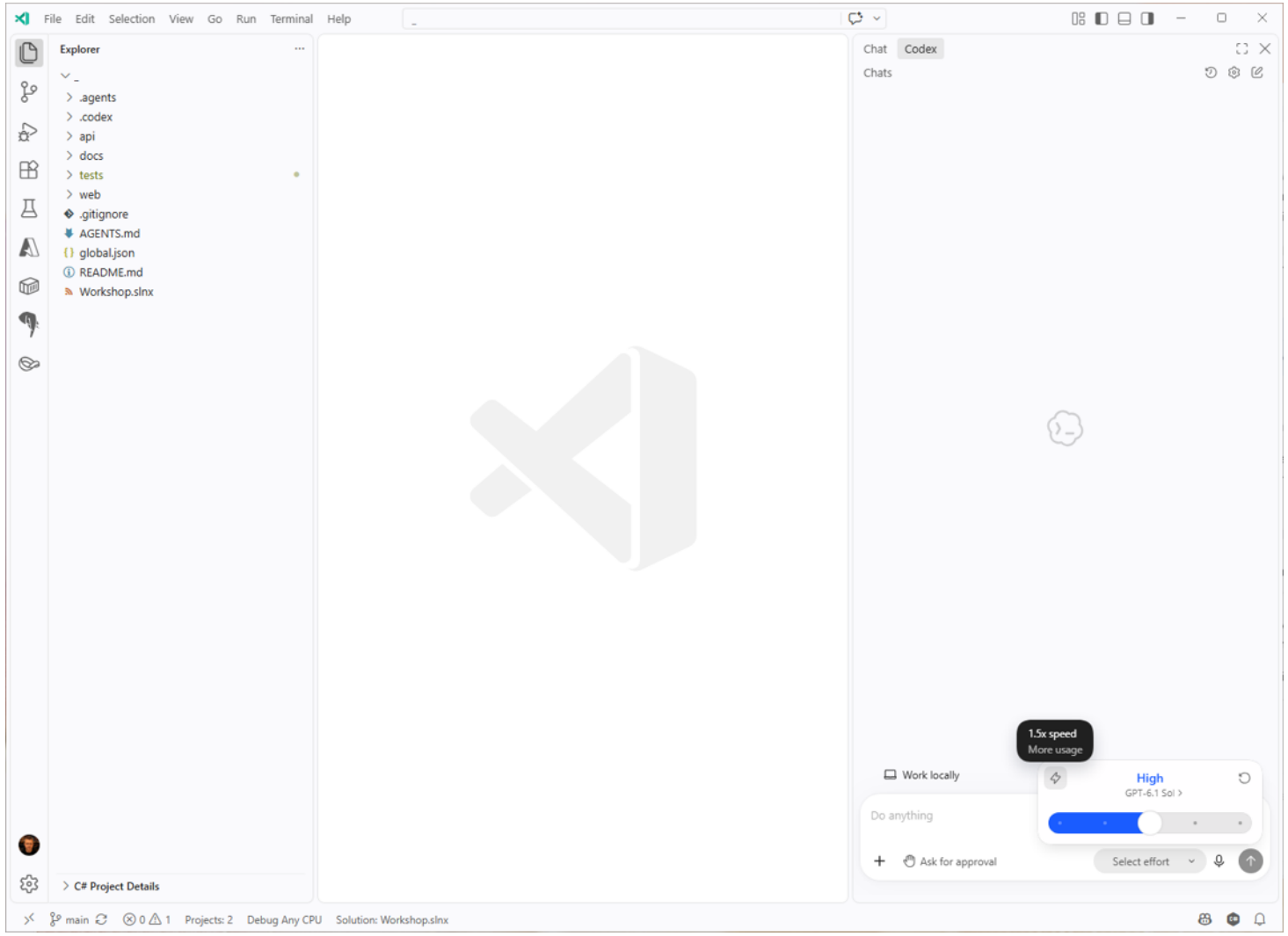

File Edit Selection View Go Run Terminal Help
Extension: Codex – OpenAI's coding agent
Codex – OpenAI's coding agent
pre-release
OpenAI openai.com | 16,531,418 (885)
Codex is a coding agent that works with you everywhere you code — included in ChatGPT Plus, Pro, Business, Edu, and Enterprise plans.
Switch to Release Version
Disable
Uninstall
Auto Update
DETAILS
FEATURES
EXTENSION PACK
Codex is OpenAI's coding agent that helps you write, review, and ship code faster. Use it side-by-side in your IDE or delegate larger tasks to the cloud.
Pair with Codex
Add Codex as a panel in VS Code to chat, edit, and preview changes seamlessly. With context from opened files and selected code, you can write shorter prompts and get faster, more relevant results. (Tip: many users like moving the Codex panel to the right side of the editor.)
Delegate to Codex in the cloud
Offload bigger jobs to Codex in the cloud, then track progress and review results without leaving your IDE. For finishing touches, open cloud tasks locally. Codex keeps the context consistent.
Sign in with your ChatGPT account
Codex works with Plus, Pro, Business, Edu, and Enterprise ChatGPT plans. Learn what's included in your plan at the Help Center.
Learn more about Codex
See the docs at https://developers.openai.com/codex/ide.
(MacOS only) Work with the ChatGPT macOS app
The OpenAI Codex extension also supports connecting the ChatGPT macOS app to VS Code so ChatGPT can answer questions or make simple edits. To use this feature, launch the ChatGPT macOS and select "Work with VS Code".
Installation
Identifier openai.chatgpt
Version 26.51021.22027
Last Updated 2 weeks ago
Size 1.54GB
Marketplace
Published 2 years ago
Last Released 2 weeks ago
Categories
AI
Chat
Resources
License
OpenAI
Marketplace
main
0 1
Projects: 2
Debug Any CPU
Solution: Workshop.slnx

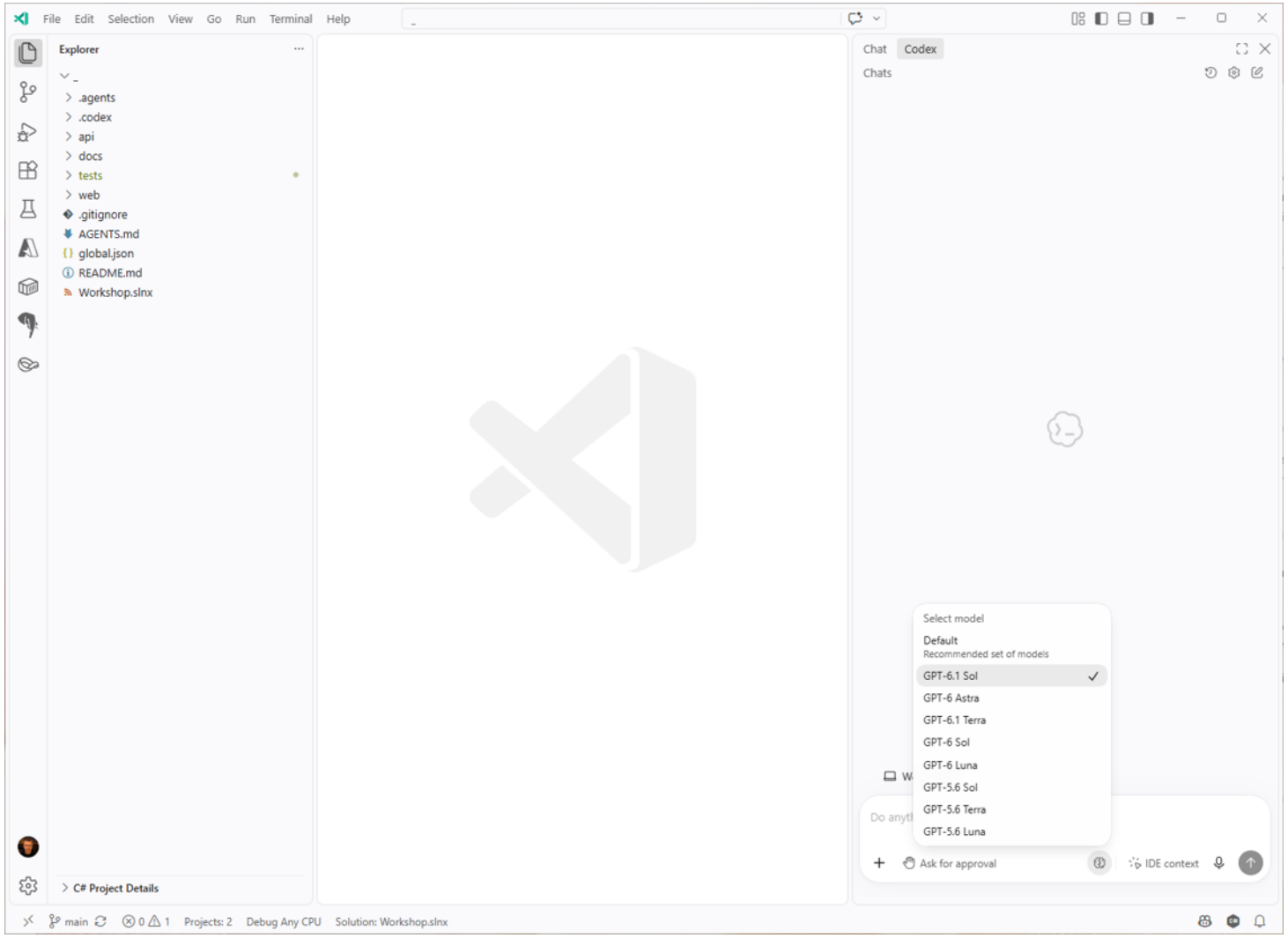

File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Chats
Work locally
Do anything
Ask for approval
IDE context
main
0 1
Projects: 2
Debug Any CPU
Solution: Workshop.slnx


File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Chats
Work in
Local
Cloud
No environment
Usage remaining
Work locally
Do anything
Ask for approval
GPT-6.1 Sol High
main
Projects: 2
Debug Any CPU
Solution: Workshop.slnx


File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Chats
Add
Files and folders
Goal Set a goal to keep pursuing
Plan mode Turn plan mode on
Sketch Draw a sketch
Plugins
Figma Create designs, ship to code
OneNote Read, export, organize, and safely update OneNote
kn-researcher Build evidence-backed briefs
kn-editorial-planner Plan ideas, series and briefs
kn-content-reviewer Review and maintain content
kn-content-writer Write knowledge content
Ask for approval
GPT-6.1 Sol High
main
Projects: 2
Debug Any CPU
Solution: Workshop.slnx

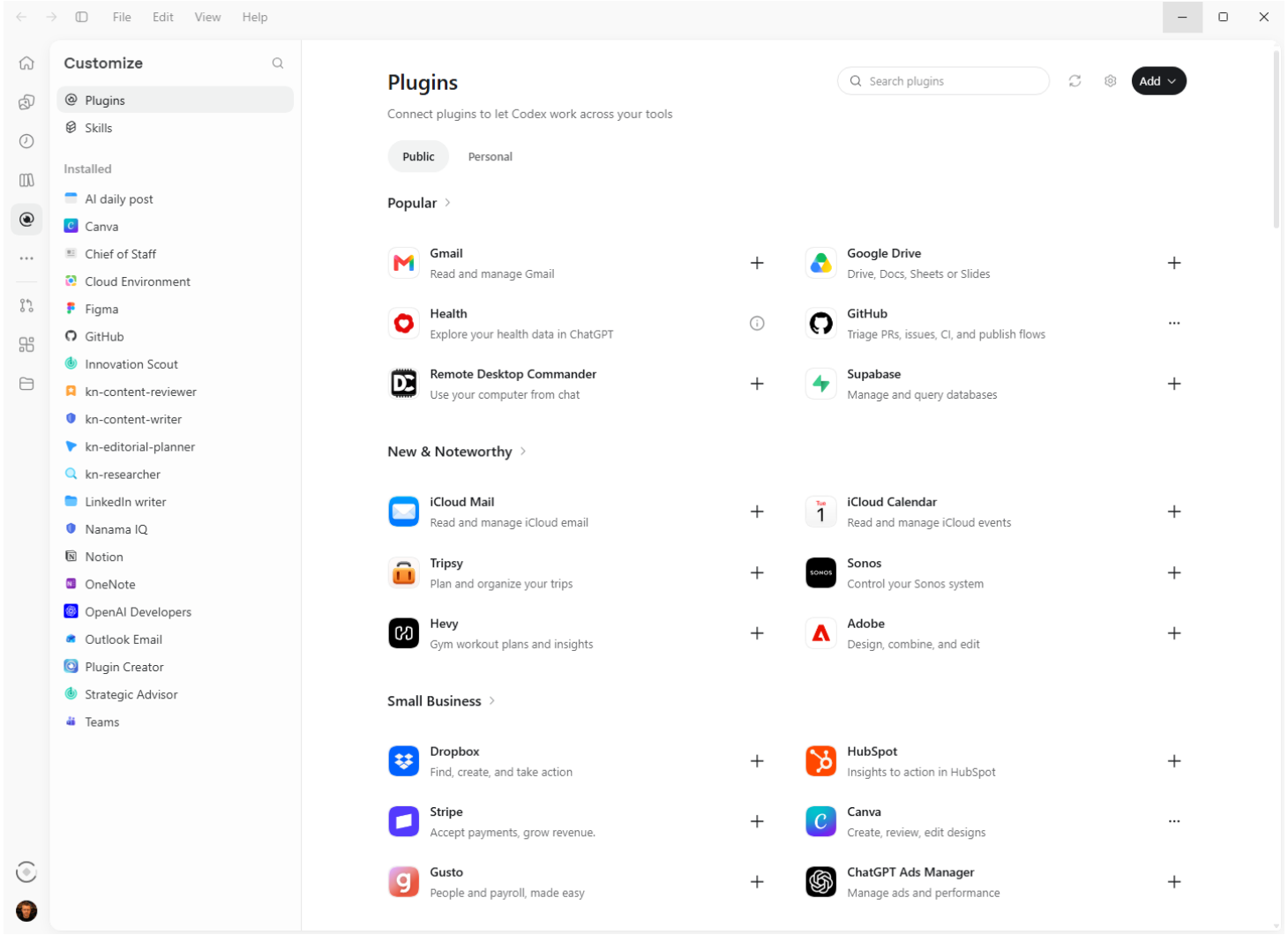

File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Chats
How should Codex actions be approved?
Learn more
Ask for approval
Always ask to edit external files and use the internet
Approve for me
Only ask for actions detected as potentially unsafe
Full access
Unrestricted access to the internet and any file on your computer
Ask for approval
IDE context
main
Projects: 2
Debug Any CPU
Solution: Workshop.slnx

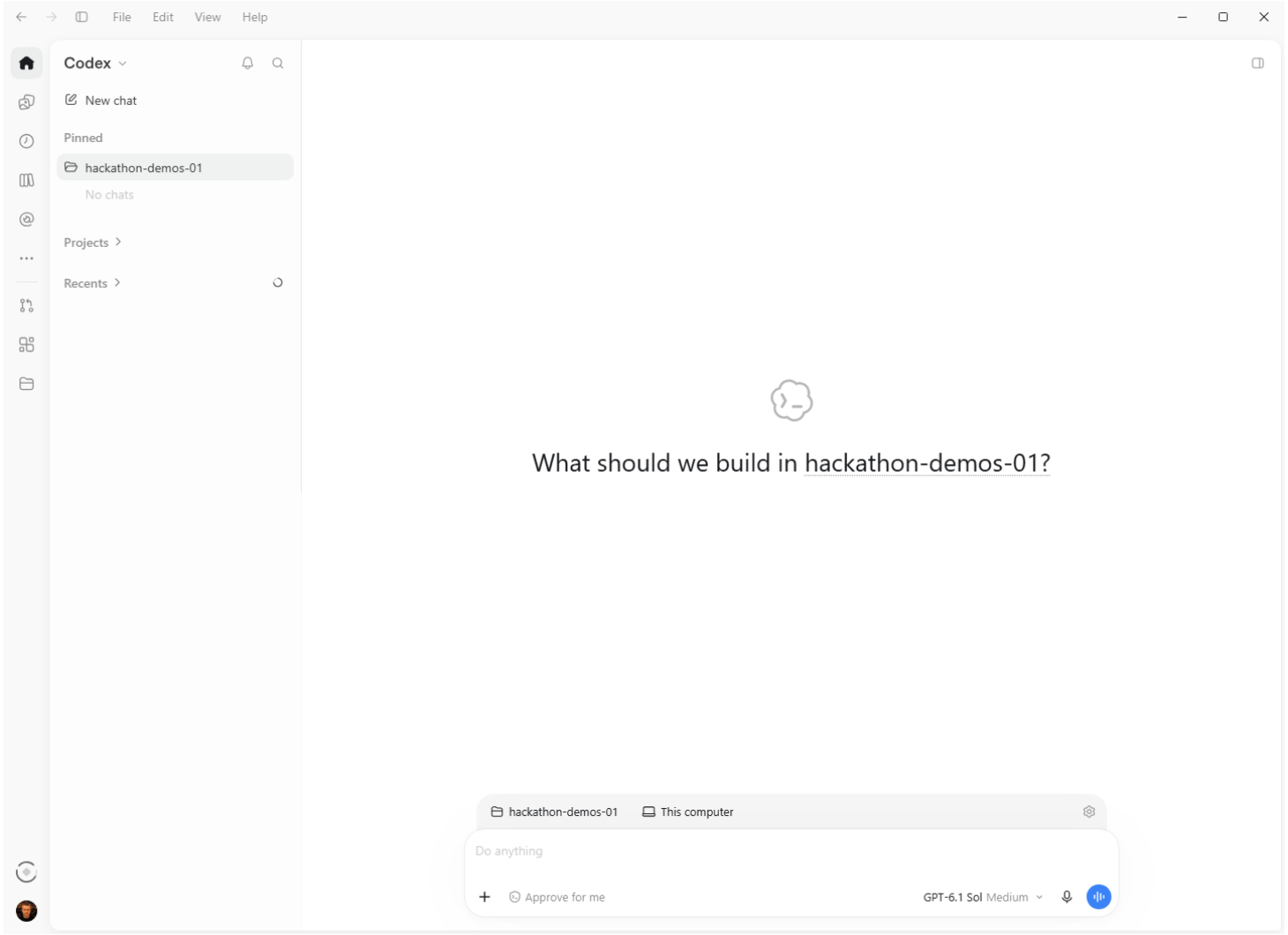

File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Chats
1.5x speed
More usage
Work locally
High
GPT-6.1 Sol
Do anything
Ask for approval
Select effort
main
Projects: 2
Debug Any CPU
Solution: Workshop.slnx

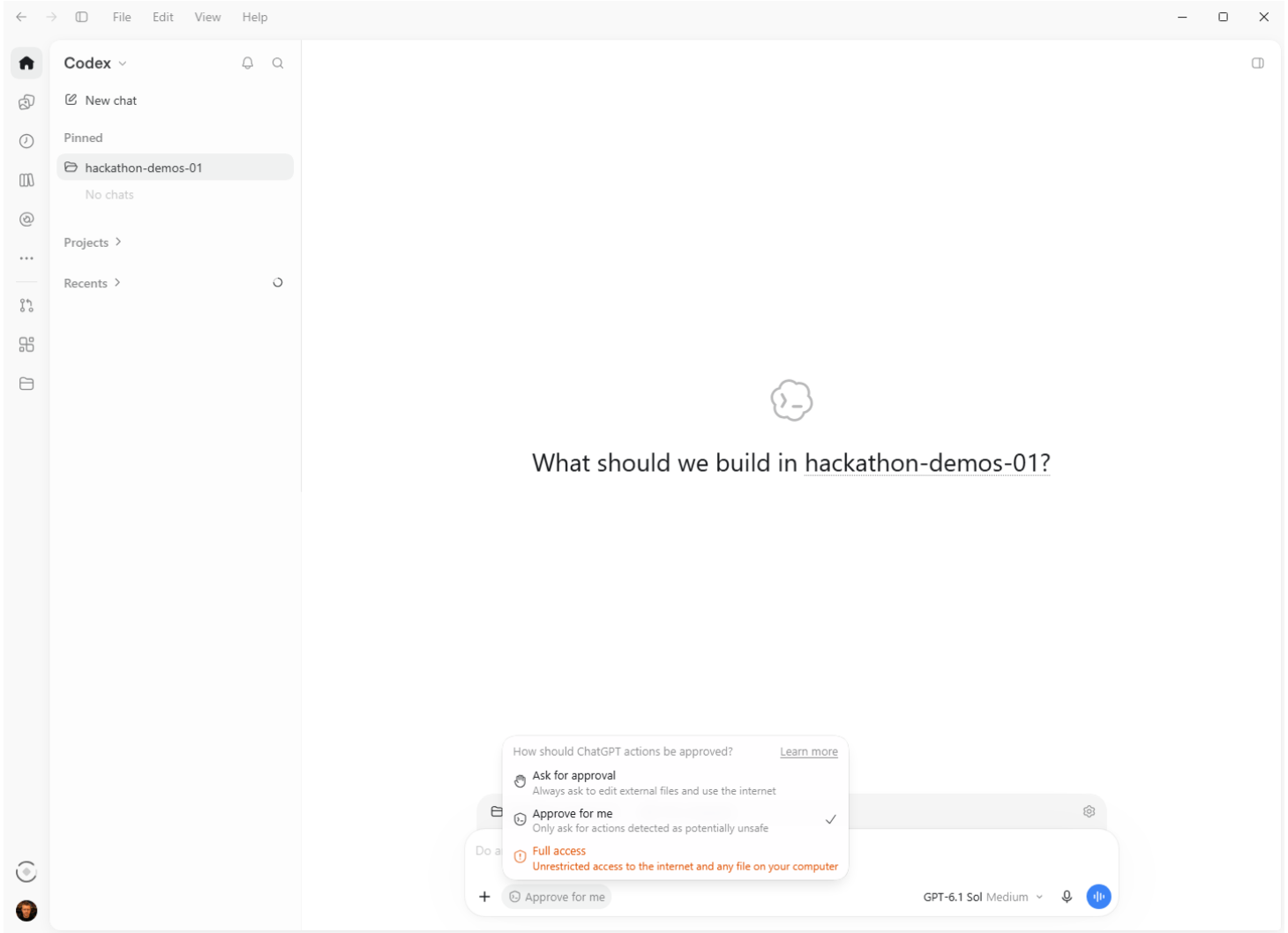

File Edit Selection View Go Run Terminal Help
Explorer
.agents
.codex
api
docs
tests
web
.gitignore
AGENTS.md
global.json
README.md
Workshop.slnx
C# Project Details
Chat
Codex
Chats
Select model
Default
Recommended set of models
GPT-6.1 Sol
GPT-6 Astra
GPT-6.1 Terra
GPT-6 Sol
GPT-6 Luna
GPT-5.6 Sol
GPT-5.6 Terra
GPT-5.6 Luna
Ask for approval
IDE context
main
Projects: 2
Debug Any CPU
Solution: Workshop.slnx

Codex desktop app

demo


File
Edit
View
Help
Customize
Plugins
Skills
Installed
AI daily post
Canva
Chief of Staff
Cloud Environment
Figma
GitHub
Innovation Scout
kn-content-reviewer
kn-content-writer
kn-editorial-planner
kn-researcher
LinkedIn writer
Nanama IQ
Notion
OneNote
OpenAI Developers
Outlook Email
Plugin Creator
Strategic Advisor
Teams
Plugins
Search plugins
Add
Connect plugins to let Codex work across your tools
Public
Personal
Popular
Gmail
Read and manage Gmail
Google Drive
Drive, Docs, Sheets or Slides
Health
Explore your health data in ChatGPT
GitHub
Triage PRs, issues, CI, and publish flows
Remote Desktop Commander
Use your computer from chat
Supabase
Manage and query databases
New & Noteworthy
iCloud Mail
Read and manage iCloud email
iCloud Calendar
Read and manage iCloud events
Tripsy
Plan and organize your trips
Sonos
Control your Sonos system
Hevy
Gym workout plans and insights
Adobe
Design, combine, and edit
Small Business
Dropbox
Find, create, and take action
HubSpot
Insights to action in HubSpot
Stripe
Accept payments, grow revenue.
Canva
Create, review, edit designs
Gusto
People and payroll, made easy
ChatGPT Ads Manager
Manage ads and performance


File
Edit
View
Help
Codex
New chat
Pinned
hackathon-demos-01
No chats
Projects
Recents
What should we build in hackathon-demos-01?
hackathon-demos-01
This computer
Do anything
Approve for me
GPT-6.1 Sol Medium


File
Edit
View
Help
Codex
New chat
Pinned
hackathon-demos-01
No chats
Projects
Recents
What should we build in hackathon-demos-01?
How should ChatGPT actions be approved?
Learn more
Ask for approval
Always ask to edit external files and use the internet
Approve for me
Only ask for actions detected as potentially unsafe
Full access
Unrestricted access to the internet and any file on your computer
Approve for me
GPT-6.1 Sol Medium